Set theory

A well defined collection of distinct objects is called a set.

- The objects are called the elements or members of the set.
- Sets are denoted by capital letters A, B, C …, X, Y, Z.
- The elements of a set are represented by lower case letters a, b, c, … , x, y, z.
- If an object x is a member of a set A, we write $x \in A$, which reads "x belongs to A" or "x is in A" or "x is an element of A", otherwise we write $x \notin A$, which reads "x does not belong to A" or "x is not in A" or "x is not an element of A".

TABULAR FORM

We list all the elements of a set, separated by commas and enclosed within braces or curly brackets{}.

EXAMPLES

In the following examples we write the sets in Tabular Form.

A = {1, 2, 3, 4, 5} is the set of first five **Natural Numbers**.
B = {2, 4, 6, 8, …, 50} is the set of **Even numbers** up to 50.
C = {1, 3, 5, 7, 9, …} is the set of **positive odd numbers.**

NOTE : The symbol "…" is called an ellipsis. It is a short for "and so forth."

DESCRIPTIVE FORM:

We state the elements of a set in words.

EXAMPLES

Now we will write the above examples in the Descriptive Form.

A = set of first five Natural Numbers. (Descriptive Form)
B = set of positive even integers less or equal to fifty. (Descriptive Form)
C = set of positive odd integers. (Descriptive Form)

SET BUILDER FORM:

We write the common characteristics in symbolic form, shared by all the elements of the set.

EXAMPLES:

Now we will write the same examples which we write in Tabular as well as Descriptive Form ,in Set Builder Form .

A = $\{x \in N \mid x \leq 5\}$ (Set Builder Form)
B = $\{x \in E \mid 0 < x \leq 50\}$ (Set Builder Form)
C = $\{x \in O \mid 0 < x\}$ (Set Builder Form)

SETS OF NUMBERS:

1. **Set of Natural Numbers**
   N = {1, 2, 3, … }
2. **Set of Whole Numbers**
   W = {0, 1, 2, 3, … }
3. **Set of Integers**
   Z = {…, -3, -2, -1, 0, +1, +2, +3, …}
   = {0, ±1, ±2, ±3, …}

{"Z" stands for the first letter of the German word for integer: Zahlen.}

4. **Set of Even Integers**
   E = {0, ± 2, ± 4, ± 6, …}
5. **Set of Odd Integers**
   O = {± 1, ± 3, ± 5, …}
6. **Set of Prime Numbers**
   P = {2, 3, 5, 7, 11, 13, 17, 19, …}
7. **Set of Rational Numbers (or Quotient of Integers)**
   Q = $\{x \mid x = \frac{p}{q},\ p, q \in Z,\ q \neq 0\}$
8. **Set of Irrational Numbers**
   $\overline{Q} = Q' = \{ x \mid x \text{ is not rational}\}$
   For example, $\sqrt{2}$, $\sqrt{3}$, π, e, etc.
9. **Set of Real Numbers**
   $R = Q \cup Q'$
10. **Set of Complex Numbers**
    C = {z | z = x + *i*y; x, y∈R} Here, $i = \sqrt{-2}$

SUBSET:

If A & B are two sets, then A is called a subset of B. It is written as A ⊆ B. The set A is subset of B **if and only if** any element of A is also an element of B.

Symbolically:

$$A \subseteq B \Leftrightarrow \text{if } x \in A, \text{ then } x \in B$$

REMARK:

1. When A ⊆ B, then B is called a superset of A.
2. When A is not subset of B, then there exist at least one x ∈ A such that x ∉B.
3. Every set is a subset of itself.

EXAMPLES:

Let

A = {1, 3, 5} B = {1, 2, 3, 4, 5}
C = {1, 2, 3, 4} D = {3, 1, 5}

Then

A ⊆ B (Because every element of A is in B)
C ⊆ B (Because every element of C is also an element of B)
A ⊆ D (Because every element of A is also an element of D)

EXAMPLE:

The set of integers "Z" is a subset of the set of Rational Number "Q", since every integer 'n' could be written as:

$$n = \frac{n}{1} \in Q$$

Hence $Z \subseteq Q$.

PROPER SUBSET:

Let A and B be sets. A is a proper subset of B, if and only if, every element of A is in B but there is at least one element of B that is not in A, and is denoted as $A \subset B$.

EXAMPLE:

Let $A = \{1, 3, 5\}$ $B = \{1, 2, 3, 5\}$

then $A \subset B$ (Because there is an element 2 of B which is not in A).

EQUAL SETS:

Two sets A and B are equal **if and only if** every element of A is in B and every element of B is in A and is denoted A = B.

Symbolically:

$$A = B \text{ iff } A \subseteq B \text{ and } B \subseteq A$$

EXAMPLE:

Let $A = \{1, 2, 3, 6\}$ B = the set of positive divisors of 6

$C = \{3, 1, 6, 2\}$ $D = \{1, 2, 2, 3, 6, 6, 6\}$

Then A, B, C, and D are all equal sets.

NULL SET:

A set which contains no element is called a **null set**, or an **empty set** or a **void set**. It is denoted by the Greek letter $\varnothing$ (phi) or { }.

EXAMPLE

$$A = \{x \mid x \text{ is a person taller than 10 feet}\} = \varnothing$$

(Because there does not exist any human being which is taller then 10 feet)

$$B = \{x \mid x^2 = 4, x \text{ is odd}\} = \varnothing$$

(Because we know that there does not exist any odd number whose square is 4)

REMARK

$\varnothing$ is regarded as a subset of every set.

EXERCISE:

Determine whether each of the following statements is true or false.

a. $x \in \{x\}$ **TRUE**

(Because x is the member of the singleton set { x })

a. $\{x\} \subseteq \{x\}$ **TRUE**

(Because Every set is the subset of itself.

Note that every Set has necessarily two subsets ∅ and the Set itself. These two subset are known as Improper subsets and any other subset is called Proper Subset)

a. $\{x\} \in \{x\}$ **FALSE**

(Because {x} is not the member of {x}) Similarly other

d. $\{x\} \in \{\{x\}\}$ **TRUE**

e. $\varnothing \subseteq \{x\}$ **TRUE**

f. $\varnothing \in \{x\}$ **FALSE**

UNIVERSAL SET:

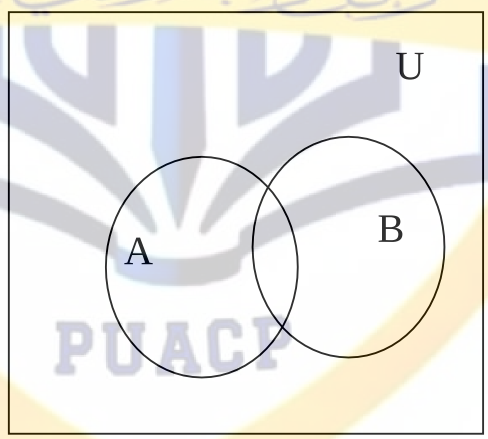
The set of all elements under consideration is called the Universal Set. The Universal Set is usually denoted by U.

Example

A = { 2, 4, 6 }
B = {1, 3, 5 }
Universal set = U = { 1, 2, 3, 4, 5, 6 }

VENN DIAGRAM:

A Venn diagram is a graphical representation of sets by regions in the plane. The Universal Set is represented by the interior of a rectangle, and the other sets are represented by disks lying within the rectangle.



FINITE AND INFINITE SETS:

A set **S** is said to be **finite** if it contains exactly *m* distinct elements where *m* denotes some non negative integer.

In such case we write $|S| = m$ or $n(S) = m$

A set is said to be **infinite** if it is not finite.

EXAMPLES:

1. The set S of letters of English alphabets is finite and $|S| = 26$
2. The null set ∅ has no elements, is finite and $|\varnothing| = 0$
3. The set of positive integers {1, 2, 3,…} is infinite.

EXERCISE:

Determine which of the following sets are finite/infinite.

1.	A = {month in the year}	**FINITE**
2.	B = {even integers}	**INFINITE**
3.	C = {positive integers less than 1}	**FINITE**
4.	D = {animals living on the earth}	**FINITE**
5.	E = {lines parallel to x-axis}	**INFINITE**
6.	$F = \{x \in \mathbf{R} \mid x^{100} + 29x^{50} - 1 = 0\}$	**FINITE**
7.	G = {circles through origin}	**INFINITE**

MEMBERSHIP TABLE:

A table displaying the membership of elements in sets. To indicate that an element is in a set, a 1 is used; to indicate that an element is not in a set, a 0 is used.

Membership tables can be used to prove set identities.

A	A^c
1	0
0	1

The above table is the Membership table for Complement of A. Now in the above table note that if an element is the member of A, then it cannot be the Member of A^c thus where in the table we have 1 for A in that row we have 0 in A^c.

Similarly, if an element is not a member of A, it will be the member of A^c.

So we have 0 for A and 1 for A^c.

Venn diagram

UNION:

Let A and B be subsets of a universal set U. The union of sets A and B is the set of all elements in U that belong to A or to B or to both, and is denoted $A \cup B$.

Symbolically:

$$A \cup B = \{x \in U \mid x \in A \text{ or } x \in B\}$$

EMAMPLE:

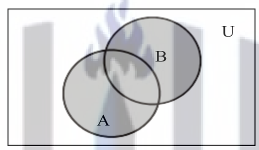
Let U = {a, b, c, d, e, f, g}

A = {a, c, e, g}, B = {d, e, f, g}

Then $A \cup B = \{x \in U \mid x \in A \text{ or } x \in B\}$

$=\{a, c, d, e, f, g\}$

VENN DIAGRAM FOR UNION:



$A \cup B$ is shaded

REMARK:

1. $A \cup B = B \cup A$ that is union is commutative you can prove this very easily only by using definition.
2. $A \subseteq A \cup B$ and $B \subseteq A \cup B$

The above remark of subset is easily seen by the definition of union.

MEMBERSHIP TABLE FOR UNION:

A	B	$A \cup B$
1	1	1
1	0	1
0	1	1
0	0	0

REMARK:

This membership table is similar to the truth table for logical connective, disjunction ($\vee$).

INTERSECTION:

Let A and B subsets of a universal set U. The intersection of sets A and B is the set of all elements in U that belong to both A and B and is denoted $A \cap B$.

Symbolically:

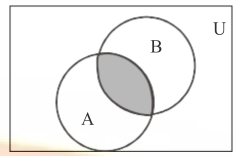
$$A \cap B = \{x \in U \mid x \in A \text{ and } x \in B\}$$

EXMAPLE:

Let $U = \{a, b, c, d, e, f, g\}$

$A = \{a, c, e, g\}$, $B = \{d, e, f, g\}$

Then $A \cap B = \{e, g\}$



$A \cap B$ is shaded

VENN DIAGRAM FOR INTERSECTION:

REMARK:

1. $A \cap B = B \cap A$
2. $A \cap B \subseteq A$ and $A \cap B \subseteq B$
3. If $A \cap B = \phi$, then A & B are called disjoint sets.

MEMBERSHIP TABLE FOR INTERSECTION:

A	B	$A \cap B$
1	1	1
1	0	0
0	1	0
0	0	0

REMARK:

This membership table is similar to the truth table for logical connective, conjunction ($\wedge$).

DIFFERENCE:

Let A and B be subsets of a universal set U. The difference of "A and B" (or relative complement of B in A) is the set of all elements in U that belong to A but not to B, and is denoted $A - B$ or $A \setminus B$.

Symbolically:

$$A - B = \{x \in U \mid x \in A \text{ and } x \notin B\}$$

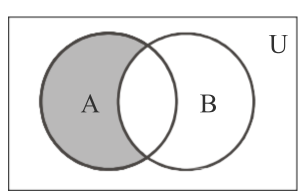
EXAMPLE:

Let $U = \{a, b, c, d, e, f, g\}$

$A = \{a, c, e, g\}$, $B = \{d, e, f, g\}$

Then $A - B = \{a, c\}$

VENN DIAGRAM FOR SET DIFFERENCE:



A-B is shaded

REMARK:

1. $A - B \neq B - A$ that is Set difference is not commutative.
2. $A - B \subseteq A$
3. $A - B$, $A \cap B$ and $B - A$ are mutually disjoint sets.

MEMBERSHIP TABLE FOR SET DIFFERENCE:

A	B	A – B
1	1	0
1	0	1
0	1	0
0	0	0

REMARK:

The membership table is similar to the truth table for $\sim (p \rightarrow q)$.

COMPLEMENT:

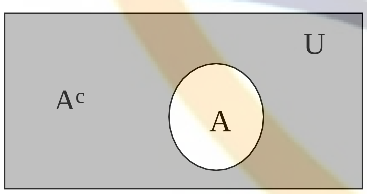
Let A be a subset of universal set U. The complement of A is the set of all element in U that do not belong to A, and is denoted AN, A or Ac
Symbolically:

$$A^c = \{x \in U \mid x \notin A\}$$

EXAMPLE:

Let $U = \{a, b, c, d, e, f, g]$
$A = \{a, c, e, g\}$
Then $A^c = \{b, d, f\}$

VENN DIAGRAM FOR COMPLEMENT:



A^c is shaded

REMARK :

1. $A^c = U - A$
2. $A \cap A^c = \phi$
3. $A \cup A^c = U$

MEMBERSHIP TABLE FOR COMPLEMENT:

A	A^c
1	0
0	1

REMARK

This membership table is similar to the truth table for logical connective negation (~)

EXERCISE:

Let $U = \{1, 2, 3, \ldots, 10\}$, $X = \{1, 2, 3, 4, 5\}$

$Y = \{y \mid y = 2x, x \in X\}$, $Z = \{z \mid z^2 - 9z + 14 = 0\}$

Enumerate:

(1) $X \cap Y$ (2) $Y \cup Z$ (3) $X - Z$

(4) Y^c (5) $X^c - Z^c$ (6) $(X - Z)^c$

Firstly we enumerate the given sets.

Given

$U = \{1, 2, 3, \ldots, 10\}$,

$X = \{1, 2, 3, 4, 5\}$

$Y = \{y \mid y = 2x, x \in X\} = \{2, 4, 6, 8, 10\}$

$Z = \{z \mid z^2 - 9z + 14 = 0\} = \{2, 7\}$

(1) $X \cap Y = \{1, 2, 3, 4, 5\} \cap \{2, 4, 6, 8, 10\}$
$= \{2, 4\}$

(2) $Y \cup Z = \{2, 4, 6, 8, 10\} \cup \{2, 7\}$
$= \{2, 4, 6, 7, 8, 10\}$

(3) $X - Z = \{1, 2, 3, 4, 5\} - \{2, 7\}$
$= \{1, 3, 4, 5\}$

(4) $Y^c = U - Y = \{1, 2, 3, \ldots, 10\} - \{2, 4, 6, 8, 10\}$
$= \{1, 3, 5, 7, 9\}$

(5) $X^c = \{6, 7, 8, 9, 10\}$
$Z^c = \{1, 3, 4, 5, 6, 8, 9, 10\}$
$X^c - Z^c = \{6, 7, 8, 9, 10\} - \{1, 3, 4, 5, 6, 8, 9, 10\}$
$= \{7\}$

(6) $(X - Z)^c = U - (X - Z)$
$= \{1, 2, 3, \ldots, 10\} - \{1, 3, 4, 5\}$
$= \{2, 6, 7, 8, 9, 10\}$

NOTE $(X - Z)^c \neq X^c - Z^c$

EXERCISE:

Given the following universal set U and its two subsets P and Q, where

$U = \{x \mid x \in Z, 0 \leq x \leq 10\}$

$P = \{x \mid x \text{ is a prime number}\}$

$Q = \{x \mid x^2 < 70\}$

(i) Draw a Venn diagram for the above

(ii) List the elements in $Pc \cap Q$

SOLUTION:

First we write the sets in Tabular form.

$U = \{x \mid x \in Z, 0 \leq x \leq 10\}$

Since it is the set of integers that are greater then or equal 0 and less or equal to 10. So we have

$U = \{0, 1, 2, 3, \ldots, 10\}$

$P = \{x \mid x \text{ is a prime number}\}$

It is the set of prime numbers between 0 and 10. Remember Prime numbers are those numbers which have only two distinct divisors.

$P = \{2, 3, 5, 7\}$

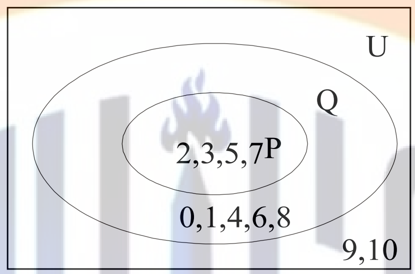
$Q = \{x \mid x^2 < 70\}$

The set Q contains the elements between 0 and 10 which have their square less or equal to 70.

$Q = \{0, 1, 2, 3, 4, 5, 6, 7, 8\}$

Thus we write the sets in Tabular form.

VENN DIAGRAM:



(i) $\mathbf{P^c \cap Q = ?}$

$P^c = U - P = \{0, 1, 2, 3, \ldots, 10\} - \{2, 3, 5, 7\}$
$= \{0, 1, 4, 6, 8, 9, 10\}$

and

$P^c \cap Q = \{0, 1, 4, 6, 8, 9, 10\} \cap \{0, 1, 2, 3, 4, 5, 6, 7, 8\}$
$= \{0, 1, 4, 6, 8\}$

EXERCISE:

Let

$U = \{1, 2, 3, 4, 5\}$, $C = \{1, 3\}$

and A and B are non empty sets. Find A in each of the following:

(i) $A \cup B = U$, $A \cap B = \phi$ and $B = \{1\}$

(ii) $A \subset B$ and $A \cup B = \{4, 5\}$

(iii) $A \cap B = \{3\}$, $A \cup B = \{2, 3, 4\}$ and $B \cup C = \{1,2,3\}$

(iv) A and B are disjoint, B and C are disjoint, and the union of A and B is the set $\{1, 2\}$.

(v)

(i) $\mathbf{A \cup B = U}$, $\mathbf{A \cap B = \phi}$ **and** $\mathbf{B = \{1\}}$

SOLUTION:

Since $A \cup B = U = \{1, 2, 3, 4, 5\}$

and $A \cap B = \phi$,

Therefore $\mathbf{A = B^c = \{1\}^c = \{2, 3, 4, 5\}}$

(ii) $\mathbf{A \subset B}$ **and** $\mathbf{A \cup B = \{4, 5\}}$ **also** $\mathbf{C = \{1, 3\}}$

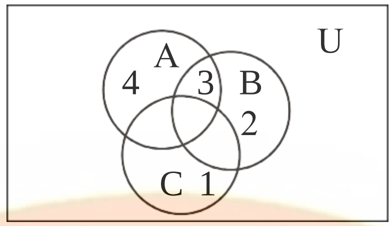
SOLUTION:

When $A \subset B$, then $A \cup B = B = \{4, 5\}$

Also A being a proper subset of B.

(iii) **$A \cap B = \{3\}$, $A \cup B = \{2, 3, 4\}$ and $B \cup C = \{1,2,3\}$**
Also $C = \{1, 3\}$

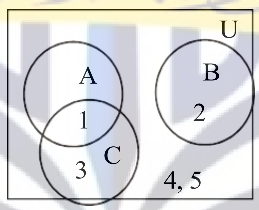
SOLUTION



Since we have 3 in the intersection of A and B as well as in C so we place 3 in common part shared by the three sets in the Venn diagram. Now since 1 is in the union of B and C it means that 1 may be in C or may be in B, but 1cannot be in B because if 1 is in the B then it must be in $A \cup B$ but 1 is not there, thus we place 1 in the part of C which is not shared by any other set. Same is the reason for 4 and we place it in the set which is not shared by any other set. Now 2 will be in B, 2 cannot be in A because $A \cap B = \{3\}$, and is not in C.
So $A = \{3, 4\}$ and $B = \{2, 3\}$

(iv) **$A \cap B = \phi$,** **$B \cap C = \phi$,** **$A \cup B = \{1, 2\}$.**
Also $C = \{1, 3\}$

SOLUTION



$A = \{1\}$

EXERCISE:

Use a Venn diagram to represent the following:

(i) $(A \cap B) \cap C^c$
(ii) $A^c \cup (B \cup C)$
(iii) $(A - B) \cap C$
(iv) $(A \cap B^c) \cup C^c$

PROVING SET IDENTITIES BY VENN DIAGRAMS:

Prove the following using Venn Diagrams:

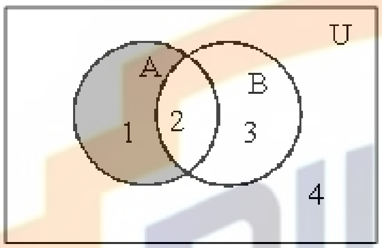
(i) $A - (A - B) = A \cap B$

(ii) $(A \cap B)^c = A^c \cup B^c$

(iii) $A - B = A \cap B^c$

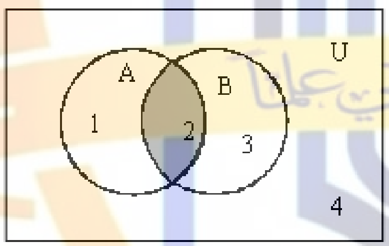
SOLUTION (i)

$$\mathbf{A - (A - B) = A \cap B}$$



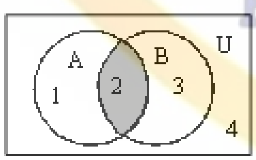
$A = \{ 1, 2 \}$
$B = \{ 2, 3 \}$
$A - B = \{ 1 \}$

$A - B$ is shaded



$A = \{ 1, 2 \}$
$A - B = \{ 1 \}$
$A - (A - B) = \{ 2 \}$

$A - (A - B)$ is shaded



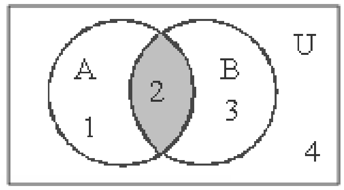
$A \cap B$ is shaded

$A = \{ 1, 2 \}$
$B = \{ 2, 3 \}$
$A \cap B = \{2\}$

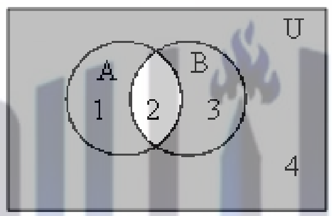
RESULT: $A - (A - B) = A \cap B$

SOLUTION (ii)

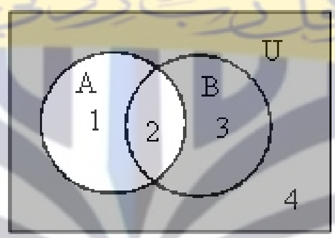
$$(A \cap B)^c = A^c \cup B^c$$



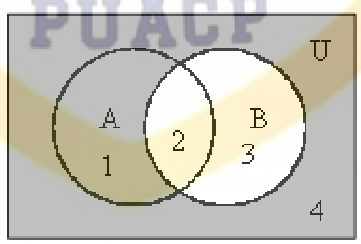
$A \cap B$



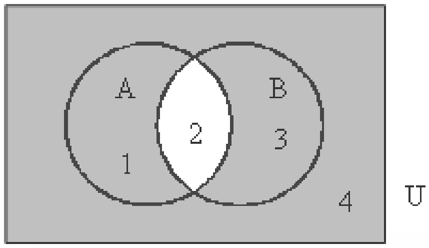
$(A \cap B)^c$ ....(a)



A^c is shaded.



B^c is shaded.



$A^c \cup B^c$ is shaded.

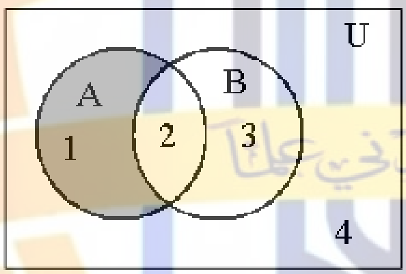
..................(b)

Now diagrams (a) and (b) are same hence

RESULT: $(A \cap B)^c = A^c \cup B^c$

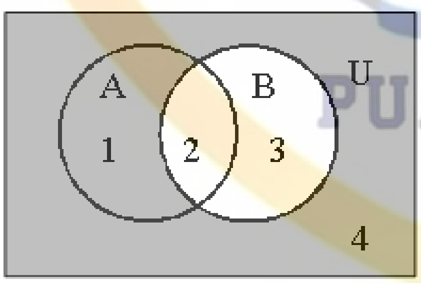
SOLUTION (iii)

$$A - B = A \cap B^c$$

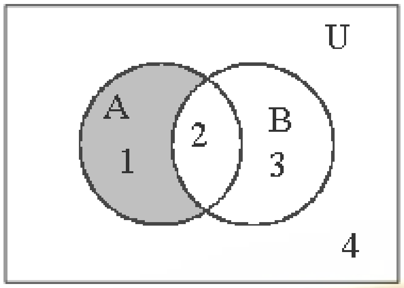


A – B is shaded.

................(a)



B^c is shaded.



$A \cap B^c$ is shaded ....(b)

From diagrams (a) and (b) we can say

RESULT: $\mathbf{A - B = A \cap B^c}$

PROVING SET IDENTITIES BY MEMBERSHIP TABLE:

Prove the following using Membership Table:

(i) $A - (A - B) = A \cap B$

(ii) $(A \cap B)^c = A^c \cup B^c$

(iii) $A - B = A \cap B^c$

SOLUTION (i)

$$\mathbf{A - (A - B) = A \cap B}$$

A	B	A-B	A-(A-B)	A∩B
1	1	0	1	1
1	0	1	0	0
0	1	0	0	0
0	0	0	0	0

Since the last two columns of the above table are same hence the corresponding set expressions are same. That is

$$\mathbf{A - (A - B) = A \cap B}$$

SOLUTION (ii)

$$\mathbf{(A \cap B)^c = A^c \cup B^c}$$

A	**B**	**A∩B**	**(A∩B)^c**	**A^c**	**B^c**	**A^c ∪ B^c**
1	**1**	**1**	**0**	**0**	**0**	**0**
1	**0**	**0**	**1**	**0**	**1**	**1**
0	**1**	**0**	**1**	**1**	**0**	**1**
0	**0**	**0**	**1**	**1**	**1**	**1**

Since the fourth and last columns of the above table are same hence the corresponding set expressions are same. That is

$$(A \cap B)^c = A^c \cup B^c$$

SOLUTION (iii)

A	B	A – B	B^c	$A \cap B^c$
1	1	0	0	0
1	0	1	1	1
0	1	0	0	0
0	0	0	1	0

Set identities

SET IDENTITIES:

Let A, B, C be subsets of a universal set U.

1. Idempotent Laws
   a. $A \cup A = A$ b. $A \cap A = A$
2. Commutative Laws
   a. $A \cup B = B \cup A$ b. $A \cap B = B \cap A$
3. Associative Laws
   a. $A \cup (B \cup C) = (A \cup B) \cup C$
   b. $A \cap (B \cap C) = (A \cap B) \cap C$
4. Distributive Laws
   a. $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$
   b. $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$
5. Identity Laws
   a. $A \cup \varnothing = A$ b. $A \cap \varnothing = \varnothing$
   c. $A \cup U = U$ d. $A \cap U = A$
6. Complement Laws
   a. $A \cup A^c = U$ b. $A \cap A^c = \varnothing$
   c. $U^c = \varnothing$ d. $\varnothing^c = U$
8. Double Complement Law
   $(A^c)^c = A$
9. DeMorgan's Laws
   a. $(A \cup B)^c = A^c \cap B^c$ b. $(A \cap B)^c = A^c \cup B^c$
10. Alternative Representation for Set Difference
    $A - B = A \cap B^c$
11. Subset Laws
    a. $A \cup B \subseteq C$ *iff* $A \subseteq C$ and $B \subseteq C$
    b. $C \subseteq A \cap B$ *iff* $C \subseteq A$ and $C \subseteq B$
12. Absorption Laws
    a. $A \cup (A \cup B) = A$ b. $A \cap (A \cup B) = A$

EXAMPLE 1:

1. **$A \subseteq A \cup B$**
2. **$A - B \subseteq A$**
3. **If $A \subseteq B$ and $B \subseteq C$ then $A \subseteq C$**
4. **$A \subseteq B$ if, and only if, $B^c \subseteq A^c$**

1. **Prove that $A \subseteq A \cup B$**

SOLUTION

Here in order to prove the identity you should remember the definition of Subset of a set. We will take the arbitrary element of a set then show that, that element is the member of the other, then the first set is the subset of the other. So

Let x be an arbitrary element of A, that is $x \in A$.

$\Rightarrow$ $x \in A$ or $x \in B$

$\Rightarrow$ $x \in A \cup B$

But x is an arbitrary element of A.

$\therefore \quad A \subseteq A \cup B$ (proved)

2. Prove that $A - B \subseteq A$

SOLUTION

Let $x \in A - B$

$\Rightarrow \quad x \in A$ and $x \notin B$ (by definition of $A - B$)

$\Rightarrow \quad x \in A$ (in particular)

But x is an arbitrary element of $A - B$

$\therefore \quad A - B \subseteq A$ (proved)

3. Prove that if $A \subseteq B$ and $B \subseteq C$, then $A \subseteq C$

SOLUTION

Suppose that $A \subseteq B$ and $B \subseteq C$

Consider $x \in A$

$\Rightarrow \quad x \in B$ (as $A \subseteq B$)

$\Rightarrow \quad x \in C$ (as $B \subseteq C$)

But x is an arbitrary element of A

$\therefore \quad A \subseteq C$ (proved)

4. Prove that $A \subseteq B$ iff $B^c \subseteq A^c$

SOLUTION:

Suppose $A \subseteq B$ {To prove $B^c \subseteq A^c$}

Let $x \in B^c$

$\Rightarrow \quad x \notin B$ (by definition of B^c)

$\Rightarrow \quad x \notin A$

$\Rightarrow \quad x \in A^c$ (by definition of A^c)

Now we know that implication and its contrapositivity are logically equivalent and the contrapositive statement of if $x \in A$ then $x \in B$ is: if $x \notin B$ then $x \notin A$ which is the definition of the $A \subseteq B$. Thus if we show for any two sets A and B, if $x \notin B$ then $x \notin A$ it means that

$A \subseteq B$. Hence

But x is an arbitrary element of B^c

$\therefore \; B^c \subseteq A^c$

Conversely,

Suppose $B^c \subseteq A^c$ {To prove $A \subseteq B$}

Let $x \in A$

$\Rightarrow \quad x \notin A^c$ (by definition of A^c)

$\Rightarrow \quad x \notin B^c$ ($\because \; B^c \subseteq A^c$)

$\Rightarrow \quad x \in B$ (by definition of B^c)

But x is an arbitrary element of A.

$\therefore \; A \subseteq B$ (proved)

EXAMPLE 2:

Let A and B be subsets of a universal set U.

Prove that $A - B = A \cap B^c$.

SOLUTION

Let $x \in A - B$

$\Rightarrow \quad x \in A$ and $x \notin B$ (definition of set difference)

$\Rightarrow$ $x \in A$ and $x \in B^c$ (definition of complement)

$\Rightarrow$ $x \in A \cap B^c$ (definition of intersection)

But x is an arbitrary element of A – B so we can write

$\therefore A - B \subseteq A \cap B^c$……………(1)

Conversely,

let $y \in A \cap B^c$

$\Rightarrow$ $y \in A$ and $y \in B^c$ (definition of intersection)

$\Rightarrow$ $y \in A$ and $y \notin B$ (definition of complement)

$\Rightarrow$ $y \in A - B$ (definition of set difference)

But y is an arbitrary element of $A \cap B^c$

$\therefore A \cap B^c \subseteq A - B$………….. (2)

From (1) and (2) it follows that

$A - B = A \cap B^c$ (as required)

EXAMPLE 3:

Prove the DeMorgan's Law: $(A \cup B)^c = A^c \cap B^c$

PROOF

Let $x \in (A \cup B)^c$

$\Rightarrow$ $x \notin A \cup B$ (definition of complement)

$x \notin A$ and $x \notin B$ (DeMorgan's Law of Logic)

$\Rightarrow$ $x \in A^c$ and $x \in B^c$ (definition of complement)

$\Rightarrow$ $x \in A^c \cap B^c$ (definition of intersection)

But x is an **arbitrary** element of $(A \cup B)^c$ so we have proved that

$\therefore (A \cup B)^c \subseteq A^c \cap B^c$………(1)

Conversely

let $y \in A^c \cap B^c$

$\Rightarrow$ $y \in A^c$ and $y \in B^c$ (definition of intersection)

$\Rightarrow$ $y \notin A$ and $y \notin B$ (definition of complement)

$\Rightarrow$ $y \notin A \cup B$ (DeMorgan's Law of Logic)

$\Rightarrow$ $y \in (A \cup B)^c$ (definition of complement)

But y is an arbitrary element of $A^c \cap B^c$

$\therefore A^c \cap B^c \subseteq (A \cup B)^c$…………………… (2)

From (1) and (2) we have

$(A \cup B)^c = A^c \cap B^c$

Which is the DeMorgan`s Law.

EXAMPLE 4:

Prove the associative law: $A \cap (B \cap C) = (A \cap B) \cap C$

PROOF:

Consider $x \in A \cap (B \cap C)$

$\Rightarrow$ $x \in A$ and $x \in B \cap C$ (definition of intersection)

$\Rightarrow$ $x \in A$ and $x \in B$ and $x \in C$ (definition of intersection)

$\Rightarrow$ $x \in A \cap B$ and $x \in C$ (definition of intersection)

$\Rightarrow$ $x \in (A \cap B) \cap C$ (definition of intersection)

But x is an arbitrary element of $A \cap (B \cap C)$

$\therefore A \cap (B \cap C) \subseteq (A \cap B) \cap C$……(1)

Conversely

let $y \in (A \cap B) \cap C$

$\Rightarrow$ $y \in A \cap B$ and $y \in C$ (definition of intersection)

$\Rightarrow$ $y \in A$ and $y \in B$ and $y \in C$ (definition of intersection)

$\Rightarrow$ $y \in A$ and $y \in B \cap C$ (definition of intersection)

$\Rightarrow$ $y \in A \cap (B \cap C)$ (definition of intersection)

But y is an arbitrary element of $(A \cap B) \cap C$

$\therefore$ $(A \cap B) \cap C \subseteq A \cap (B \cap C)$… .....(2)

From (1) & (2), we conclude that

$A \cap (B \cap C) = (A \cap B) \cap C$ (proved)

EXAMPLE 5:

For any sets A and B if $A \subseteq B$ then $A \cap B = A$

SOLUTION:

Let $x \in A \cap B$

$\Rightarrow$ $x \in A$ and $x \in B$

$\Rightarrow$ $x \in A$ (in particular)

Hence $A \cap B \subseteq A$… ....(1)

Conversely,

let $x \in A$.

Then $x \in B$ (since $A \subseteq B$)

Now $x \in A$ and $x \in B$, therefore $x \in A \cap B$

Hence, $A \subseteq A \cap B$… ....(2)

From (1) and (2) it follows that

$A = A \cap B$ (proved)

USING SET IDENTITIES:

For all subsets A and B of a universal set U, prove that

$$(A - B) \cup (A \cap B) = A$$

PROOF:

LHS $= (A - B) \cup (A \cap B)$

$= (A \cap B^c) \cup (A \cap B)$ (Alternative representation for set difference)

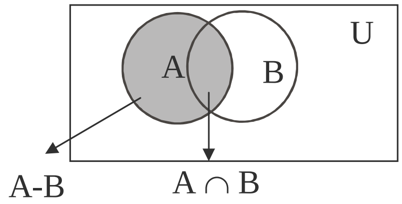
$= A \cap (B^c \cup B)$ Distributive Law

$= A \cap U$ Complement Law

$= A$ Identity Law

$=$ RHS (proved)

The result can also be seen by Venn diagram.



EXAMPLE 6:

For any two sets A and B prove that $A - (A - B) = A \cap B$

SOLUTION

$$
\begin{aligned}
\text{LHS} &= A - (A - B) \\
&= A - (A \cap B^c) && \text{Alternative representation for set difference} \\
&= A \cap (A \cap B^c)^c && \text{Alternative representation for set difference} \\
&= A \cap (A^c \cup (B^c)^c) && \text{DeMorgan's Law} \\
&= A \cap (A^c \cup B) && \text{Double Complement Law} \\
&= (A \cap A^c) \cup (A \cap B) && \text{Distributive Law} \\
&= \varnothing \cup (A \cap B) && \text{Complement Law} \\
&= A \cap B && \text{Identity Law} \\
&= \text{RHS} && \text{(proved)}
\end{aligned}
$$

EXAMPLE 7:

For all set A, B, and C prove that $(A - B) - C = (A - C) - B$

SOLUTION

$$
\begin{aligned}
\text{LHS} &= (A - B) - C \\
&= (A \cap B^c) - C && \text{Alternative representation of set difference} \\
&= (A \cap B^c) \cap C^c && \text{Alternative representation of set difference} \\
&= A \cap (B^c \cap C^c) && \text{Associative Law} \\
&= A \cap (C^c \cap B^c) && \text{Commutative Law} \\
&= (A \cap C^c) \cap B^c && \text{Associative Law} \\
&= (A - C) \cap B^c && \text{Alternative representation of set difference} \\
&= (A - C) - B && \text{Alternative representation of set difference} \\
&= \text{RHS} && \text{(proved)}
\end{aligned}
$$

EXAMPLE 8:

Simplify $(B^c \cup (B^c - A))^c$

SOLUTION

$(B^c \cup (B^c - A))^c = (B^c \cup (B^c \cap A^c))^c$

Alternative representation for set difference

$$
\begin{aligned}
&= (B^c)^c \cap (B^c \cap A^c)^c && \text{DeMorgan's Law} \\
&= B \cap ((B^c)^c \cup (A^c)^c) && \text{DeMorgan's Law} \\
&= B \cap (B \cup A) && \text{Double Complement Law} \\
&= B && \text{Absorption Law}
\end{aligned}
$$

which is the simplified form of the given expression.

PROVING SET IDENTITIES BY MEMBERSHIP TABLE:

Prove the following using Membership Table:

(i) $A - (A - B) = A \cap B$

(ii) $(A \cap B)^c = A^c \cup B^c$
(iii) $A - B = A \cap B^c$

Solution (i): $A - (A - B) = A \cap B$

A	B	A-B	A-(A-B)	$A \cap B$
1	1	0	1	1
1	0	1	0	0
0	1	0	0	0
0	0	0	0	0

Solution (ii): $(A \cap B)^c = A^c \cup B^c$

A	**B**	**$A \cap B$**	**$(A \cap B)^c$**	**A^c**	**B^c**	**$A^c \cup B^c$**
1	1	1	0	0	0	0
1	0	0	1	0	1	1
0	1	0	1	1	0	1
0	0	0	1	1	1	1

Solution (iii): $A - B = A \cap B^c$

A	B	A – B	B^c	$A \cap B^c$
1	1	0	0	0
1	0	1	1	1
0	1	0	0	0
0	0	0	1	0

Applications of Venn diagram

Exercise: A number of computer users are surveyed to find out if they have a printer, modem or scanner. Draw separate Venn diagrams and shade the areas, which represent the following configurations.

1. modem and printer but no scanner
2. scanner but no printer and no modem
3. scanner or printer but no modem.
4. no modem and no printer.

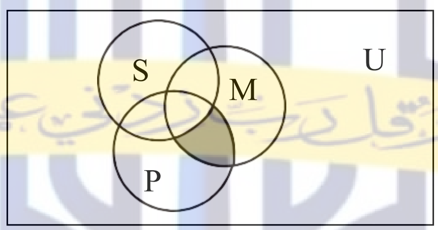
SOLUTION

Let

P represent the set of computer users having printer.
M represent the set of computer users having modem.
S represent the set of computer users having scanner.

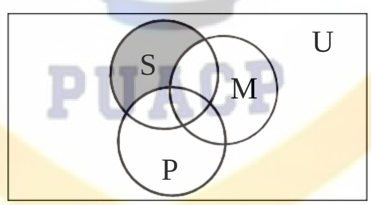
SOLUTION (i)

Modem and printer but no Scanner is shaded.



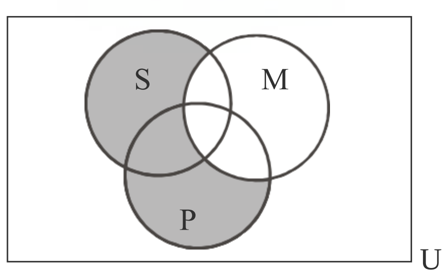
SOLUTION (ii)

Scanner but no printer and no modem is shaded.

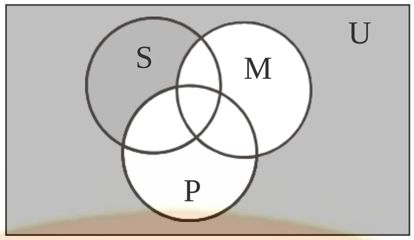


SOLUTION (iii)

Scanner or printer but no modem is shaded.



SOLUTION (iv)
No modem and no printer is shaded.

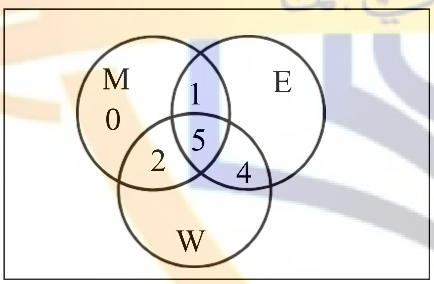


EXERCISE:

Of **21 typists** in an office, **5** use all **manual typewriters (M), electronic typewriters (E) and word processors (W); 9** use **E** and **W**; **7** use **M** and **W**; **6** use **M** and **E**; but **no one** uses **M** only.

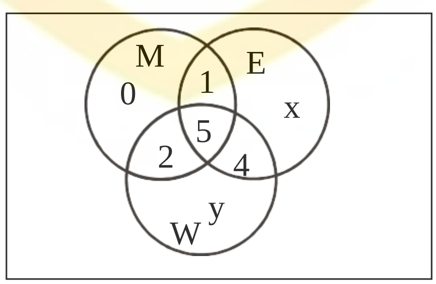
(i) Represent this information in a Venn Diagram.
(ii) If the same number of typists use electronic as use word processors, then
1. (a) How many use word processors only,
2. (b) How many use electronic typewriters?

SOLUTION (i)



SOLUTION (ii-a)

Let the number of typists using electronic typewriters (E) only be x, and the number of typists using word processors (W) only be y.



Total number of typists using E = Total Number of typists using W

$$1 + 5 + 4 + x = 2 + 5 + 4 + y$$

or, $x - y = 1$ ....(1)

Also, total number of typists = 21

$$\Rightarrow 0 + x + y + 1 + 2 + 4 + 5 = 21$$

or, $x + y = 9$..(2)
Solving (1) & (2), we get

$$x = 5, \; y = 4$$

$\therefore$ Number of typists using word processor only is $y = 4$

(ii)-(b) How many typists use electronic typewriters?

SOLUTION:

Typists using electronic typewriters = No. of elements in E

$$= 1 + 5 + 4 + x$$
$$= 1 + 5 + 4 + 5$$
$$= 15$$

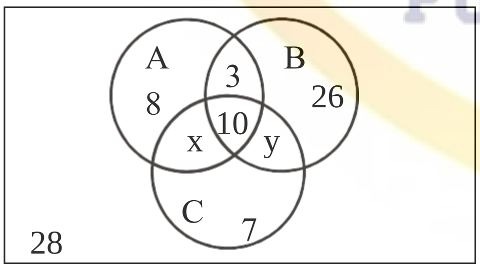
EXERCISE

In a school, 100 students have access to three software packages, A, B and C
28 did not use any software
8 used only packages A
26 used only packages B
7 used only packages C
10 used all three packages
13 used both A and B

(i) Draw a Venn diagram with all sets enumerated as for as possible. Label the two subsets which cannot be enumerated as x and y, in any order.
(ii) If twice as many students used package B as package A, write down a pair of simultaneous equations in x and y.
(iii) Solve these equations to find x and y.
(iv) How many students used package C?

SOLUTION(i)

Venn Diagram with all sets enumerated.



(ii) If twice as many students used package B as package A, write down a pair of simultaneous equations in x and y.

SOLUTION(ii):

We are given that

Number of students using package B = 2 (Number of students using package A)

Now the number of students which used package B and A are clear from the diagrams given below. So we have the following equation

$$\Rightarrow\ 3 + 10 + 26 + y = 2(8 + 3 + 10 + x)$$
$$\Rightarrow\ 39 + y = 42 + 2x$$
$$\text{or}\quad y = 2x + 3 \ldots\ldots\ldots\ldots(1)$$

Also, total number of students = 100.

Hence, $8 + 3 + 26 + 10 + 7 + 28 + x + y = 100$

or $82 + x + y = 100$

or $x + y = 18 \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots (2)$

(iii) Solving simultaneous equations for x and y.

SOLUTION(iii):

$$y = 2x + 3 \ldots\ldots\ldots\ldots\ldots\ldots(1)$$
$$x + y = 18 \ldots\ldots\ldots\ldots\ldots\ldots(2)$$

Using (1) in (2), we get,

$$x + (2x + 3) = 18$$
$$\text{or}\quad 3x + 3 = 18$$
$$\text{or}\quad 3x = 15$$
$$x = 5$$

Consequently $y = 13$

How many students used package C?

SOLUTION (iv) :

No. of students using package C

$$= x + y + 10 + 7$$
$$= 5 + 13 + 10 + 7$$
$$= 35$$

EXAMPLE:

Use diagrams to show the validity of the following argument:

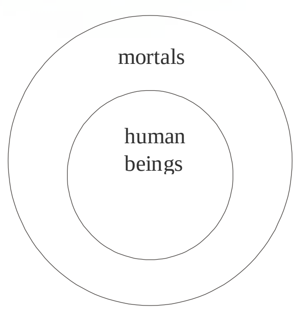
All human beings are mortal

Zeus is not mortal

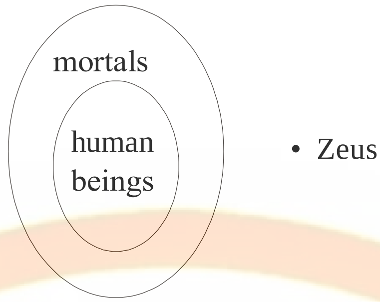
∴ Zeus is not a human being

SOLUTION:

The premise "All human beings are mortal is pictured by placing a disk labeled "human beings" inside a disk labeled "mortals". We place the disk of human Beings in side the Disk of mortals because there are things which are mortal but not Human beings so the set of human beings is subset of set of Mortals.



The second premise “Zeus is not mortal” could be pictured by placing a dot labeled “Zeus” outside the disk labeled “mortals”



Argument is valid.

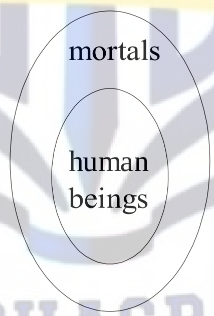
EXAMPLE:

Use a diagram to show the invalidity of the following argument:

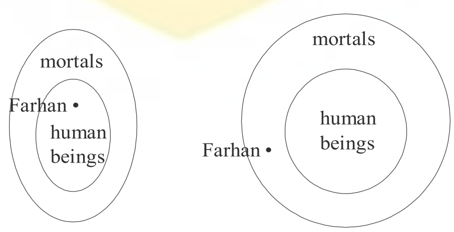
All human beings are mortal.
Farhan is mortal
∴ Farhan is a human being

SOLUTION:

The first premise “All human beings are mortal” is pictured as:



The second premise “Farhan is mortal” is represented by a dot labeled “Farhan” inside the mortal disk in either of the following two ways:



argument is invalid.

EXAMPLE

Use diagrams to test the following argument for validity:

No polynomial functions have horizontal asymptotes.

This function has a horizontal asymptote.
∴ This function is not polynomial.

SOLUTION

The premise "No polynomial functions have horizontal asymptotes" can be represented diagrammatically by two**disjoint** disks labeled "polynomial functions" and "functions with horizontal tangents.

The argument is valid.

EXERCISE:

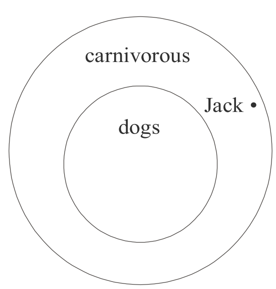
Use a diagram to show that the following argument can have true premises and a false conclusion.

All dogs are carnivorous.
Jack is not a dog.
∴ Jack is not carnivorous

SOLUTION:

The premise "All dogs are carnivorous" is pictured by placing a disk labeled "dogs" inside a disk labeled "carnivorous". :

The second premise "Jack is not a dog" could be represented by placing a dot outside the disk labeled "dogs" but inside the disk labeled "carnivorous" to make the conclusion "Jack is not carnivorous" false.



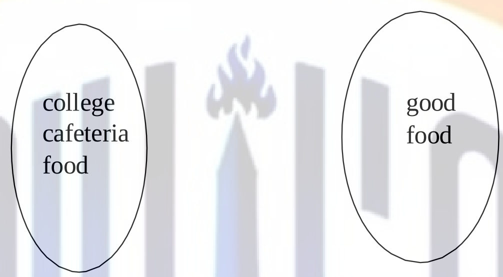
EXERCISE:

Indicate by drawing diagrams, whether the argument is valid or invalid.

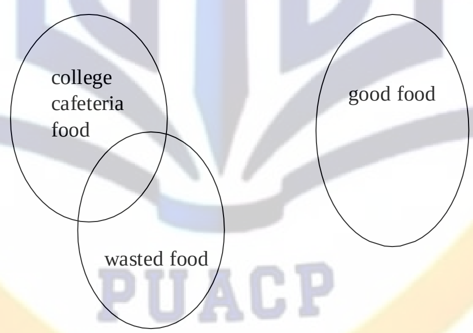
No college cafeteria food is good.
No good food is wasted.
∴ No college cafeteria food is wasted.

SOLUTION

The premise "No college food is good" could be represented by two disjoint disks shown below.



The next premise "No good food is wasted" introduces another disk labeled "wasted food" that does not overlap the disk labeled "good food", but may intersect with the disk labeled "college cafeteria food."



Argument is **invalid**

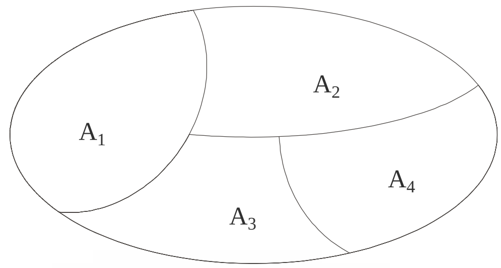
PARTITION OF A SET

A set may be divided up into its disjoint subsets. Such division is called a partition.

More precisely,

A partition of a set **A** is a collection of non- empty subsets $\{A_1, A_2, \ldots A_n\}$ of A, such that

1. $A = A_1 \cup, A_2 \cup \ldots \cup A_n$
2. $A_1, A_2, \ldots, A_n$ are mutually disjoint (or pair wise disjoint),
   i.e., $\forall$ i, j = 1, 2, …, n $A_i \cap A_j = \varnothing$ whenever $i \neq j$



A partition of a set

<u>POWER SET:</u>

The power set of a set A is the set of all subsets of A, denoted $P(A)$.

<u>EXAMPLE</u>:

Let A = {1, 2}, then
$P(A) = \{\varnothing, \{1\}, \{2\}, \{1, 2\}\}$

<u>REMARK</u>:

If A has n elements then $P(A)$ has 2^n elements.

<u>EXERCISE</u>

a. Find $P(\varnothing)$ b. Find $P(P(\varnothing))$ c. Find $P(P(P(\varnothing)))$

<u>SOLUTION:</u>

a. Since $\varnothing$ contains no element, therefore $P(\varnothing)$ will contain $2^0 =1$ element.
$P(\varnothing) = \{\varnothing\}$

b. Since $P(\varnothing)$ contains one element, namely ϕ, therefore $P(\varnothing)$ will contain $2^1 = 2$ elements
$P(P(\varnothing)) = \{\varnothing,\{\varnothing\}\}$

c. Since $P(P(\varnothing))$ contains two elements, namely $\varnothing$ and $\{\varnothing\}$, so $P(P(P(\varnothing)))$ will contain $2^2 = 4$ elements.
$P(P(P(\varnothing)))= \{\varnothing,\{\varnothing\},\{\{\varnothing\}\}, \{\varnothing,\{\varnothing\}\}\}$

Relations

ORDERED PAIR:

An ordered pair (a, b) consists of two elements "a" and "b" in which "a" is the first element and "b" is the second element.

The ordered pairs (a, b) and (c, d) are equal if, and only if, a= c and b = d.

Note that (a, b) and (b, a) are not equal unless a = b.

EXERCISE:

Find x and y given (2x, x + y) = (6, 2)

SOLUTION:

Two ordered pairs are equal if and only if the corresponding components are equal. Hence, we obtain the equations:

$2x = 6$ .... (1)

and $x + y = 2$ ....(2)

Solving equation (1) we get x = 3 and when substituted in (2) we get y = -1.

ORDERED n-TUPLE:

The ordered n-tuple $(a_1, a_2, \ldots, a_n)$ consists of elements $a_1, a_2, \ldots, a_n$ together with the ordering: first a_1, second a_2, and so forth up to a_n. In particular, an ordered 2-tuple is called an ordered pair, and an ordered 3-tuple is called an ordered triple.

Two ordered n-tuples $(a_1, a_2, \ldots, a_n)$ and $(b_1, b_2, \ldots, b_n)$ are equal if and only if each corresponding pair of their elements is equal, i.e., $a_i = b_j$, for all i, j = 1, 2, …, n.

CARTESIAN PRODUCT OF TWO SETS:

Let A and B be sets. The Cartesian product of A and B, denoted by A × B (read as "A cross B") is the set of all ordered pairs (a, b), where a is in A and b is in B.

Symbolically:

$$A \times B = \{(a, b) | a \in A \text{ and } b \in B\}$$

NOTE: If set A has ***m*** elements and set B has ***n*** elements then A ×B has ***m*** × ***n*** elements.

EXAMPLE:

Let A = {1, 2}, B = {a, b, c} then

A × B = {(1, a), (1, b), (1, c), (2, a), (2, b), (2, c)}

B × A = {(a, 1), (a, 2), (b, 1), (b, 2), (c, 1), (c, 2)}

A × A = {(1, 1), (1,2), (2, 1), (2, 2)}

B × B = {(a, a), (a, b), (a, c), (b, a), (b, b), (b, c), (c, a), (c, b),(c, c)}

REMARK:

1. $A \times B \neq B \times A$ for non-empty and unequal sets A and B.
2. $A \times \phi = \phi \times A = \phi$
3. $|A \times B| = |A| \times |B|$

CARTESIAN PRODUCT OF MORE THAN TWO SETS:

The Cartesian product of sets $A_1, A_2, \ldots, A_n$, denoted $A_1 \times A_2 \times \ldots \times A_n$, is the set of all ordered n-tuples $(a_1, a_2, \ldots, a_n)$ where $a_1 \in A_1$, $a_2 \in A_2, \ldots, a_n \in A_n$.

Symbolically:

$$A_1 \times A_2 \times \ldots \times A_n = \{(a_1, a_2, \ldots, a_n) \mid a_i \in A_i, \text{ for } i = 1, 2, \ldots, n\}$$

BINARY RELATION:

Let A and B be sets. The binary relation R from A to B is a subset of $A \times B$.

When $(a, b) \in R$, we say 'a' is related to 'b' by R, written aRb.

Otherwise, if $(a, b) \notin R$, we write $a \underline{R}\, b$.

EXAMPLE:

Let $A = \{1, 2\}$, $B = \{1, 2, 3\}$

Then $A \times B = \{(1, 1), (1, 2), (1, 3), (2, 1), (2, 2), (2, 3)\}$

Let

$R_1 = \{(1,1), (1, 3), (2, 2)\}$

$R_2 = \{(1, 2), (2, 1), (2, 2), (2, 3)\}$

$R_3 = \{(1, 1)\}$

$R_4 = A \times B$

$R_5 = \varnothing$

All being subsets of $A \times B$ are relations from A to B.

DOMAIN OF A RELATION:

The domain of a relation R from A to B is the set of all first elements of the ordered pairs which belong to R denoted by Dom(R).

Symbolically,

$$\text{Dom}(R) = \{a \in A \mid (a, b) \in R\}$$

RANGE OF A RELATION:

The range of a relation R from A to B is the set of all second elements of the ordered pairs which belong to R denoted Ran(R).

Symbolically,

$$\text{Ran}(R) = \{b \in B \mid (a, b) \in R\}$$

EXERCISE:

Let $A = \{1, 2\}$, $B = \{1, 2, 3\}$,

Define a binary relation R from A to B as follows:

$$R = \{(a, b) \in A \times B \mid a < b\}$$

Then

a. Find the ordered pairs in R.
b. Find the Domain and Range of R.
c. Is 1R3, 2R2?

SOLUTION:

Given $A = \{1, 2\}$, $B = \{1, 2, 3\}$,

$A \times B = \{(1,1), (1,2), (1,3), (2,1), (2,2), (2,3)\}$

a. $R = \{(a, b) \in A \times B \mid a < b\}$

$R = \{(1,2), (1,3), (2,3)\}$

b. $Dom(R) = \{1,2\}$ and $Ran(R) = \{2, 3\}$

c. Since $(1, 3) \in R$ so $1R3$

But $(2, 2) \notin R$ so 2 is not related with3 or $2 \,\underline{R}\, 2$

EXAMPLE:

Let A = {eggs, milk, corn} and B = {cows, goats, hens}
Define a relation R from A to B by $(a, b) \in R$ *iff* a is produced by b.
Then R = {(eggs, hens), (milk, cows), (milk, goats)}
Thus, with respect to this relation eggs R hens , milk R cows, etc.

EXERCISE :

Find all binary relations from $\{0,1\}$ to $\{1\}$

SOLUTION:

Let $A = \{0,1\}$ & $B = \{1\}$
Then $A \times B = \{(0,1), (1,1)\}$
All binary relations from A to B are in fact all subsets of $A \times B$, which are:

$R_1 = \varnothing$
$R_2 = \{(0,1)\}$
$R_3 = \{(1,1)\}$
$R_4 = \{(0,1), (1,1)\} = A \times B$

REMARK:

If $|A| = m$ and $|B| = n$
Then as we know that the number of elements in $A \times B$ are $m \times n$. Now as we know that the total number of and the total number of relations from A to B are $2^{m \times n}$.

RELATION ON A SET:

A relation on the set A is a relation from A to A.
In other words, a relation on a set A is a subset of $A \times A$.

EXAMPLE:

Let $A = \{1, 2, 3, 4\}$
Define a relation R on A as
$(a,b) \in R$ *iff* a divides b {symbolically written as $a \mid b$}
Then $R = \{(1,1), (1,2), (1,3), (1,4), (2,2), (2,4), (3,3), (4,4)\}$

REMARK:

For any set A

1. $A \times A$ is known as the universal relation.
2. $\varnothing$ is known as the empty relation.

EXERCISE:

Define a binary relation E on the set of the integers Z, as follows:

for all m,n $\in$Z, m E n $\Leftrightarrow$ m – n is even.

a. Is 0E0? Is 5E2? Is (6,6) $\in$E? Is (-1,7) $\in$E?

b. Prove that for any even integer n, nE0.

SOLUTION

E = {(m, n) $\in$Z $\times$Z | m – n is even}

a. (i) (0,0) $\in$ Z $\times$Z and 0-0 = 0 is even

Therefore 0E0.

(ii) Since (5,2)$\in$Z $\times$Z but 5-2 = 3 is not even

so $5\,\underline{E}\,2$

(iii) (6,6) $\in$ E since 6-6 = 0 is an even integer.

(iv) (-1,7) $\in$E since (-1) – 7 = -8 is an even integer.

b. For any even integer, n, we have

n – 0 = n, an even integer

so (n, 0) $\in$E or equivalently n E 0

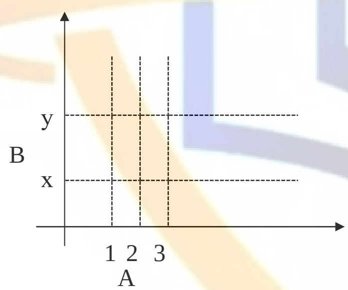
COORDINATE DIAGRAM (GRAPH) OF A RELATION:

Let A = {1, 2, 3} and B = {x, y}

Let R be a relation from A to B defined as

R = {(1, y), (2, x), (2, y), (3, x)}

The relation may be represented in a coordinate diagram as follows:



EXAMPLE:

Draw the graph of the binary relation C from R to R defined as follows:

for all (x, y) $\in$R $\times$ R, (x, y) $\in$C $\Leftrightarrow$ $x^2 + y^2 = 1$

SOLUTION

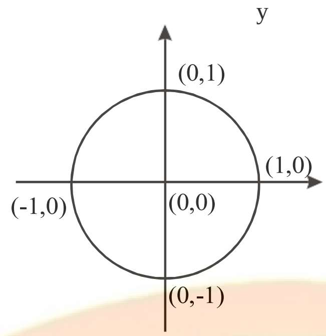
All ordered pairs (x, y) in relation C satisfies the equation $x^2+y^2=1$, which when solved for y gives:

Clearly y is real, whenever $-1 \le x \le 1$

Similarly x is real, whenever $-1 \le y \le 1$

Hence the graph is limited in the range $-1 \le x \le 1$ and $-1 \le y \le 1$

The graph of relation is



ARROW DIAGRAM OF A RELATION:

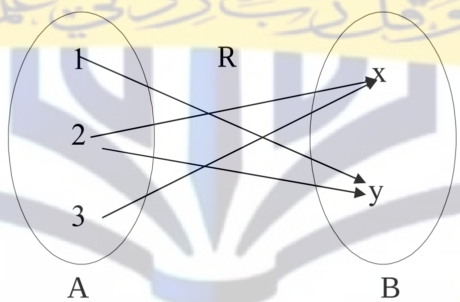
Let

A = {1, 2, 3}, B = {x, y} and

R = {1,y), (2,x), (2,y), (3,x)}

be a relation from A to B.

The arrow diagram of R is:

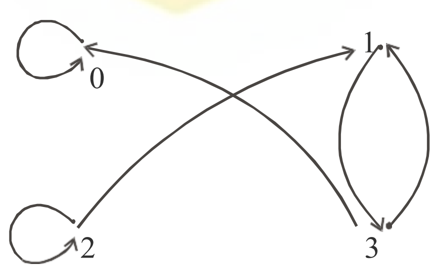


DIRECTED GRAPH OF A RELATION:

Let A = {0, 1, 2, 3}

and R = {(0,0), (1,3), (2,1), (2,2), (3,0), (3,1)}

be a binary relation on A.



DIRECTED GRAPH

MATRIX REPRESENTATION OF A RELATION

Let A = {a_1, a_2, …, a_n} and B = {b_1, b_2, …, b_m}. Let R be a relation from A to B. Define the n × m order matrix M by

$$m(i, j) = \begin{cases} 1 \text{ if } & (a_i, b_i) \in R \\ 0 \text{ if } & (a_i, b_i) \notin R \end{cases}$$

for i=1,2,…,n and j=1,2,…,m

EXAMPLE:

Let A = {1, 2, 3} and B = {x, y}
Let R be a relation from A to B defined as
R ={(1,y), (2,x), (2,y), (3,x)}

$$M = \begin{array}{c} \\ 1 \\ 2 \\ 3 \end{array} \begin{array}{c} \begin{array}{cc} x & y \end{array} \\ \begin{bmatrix} 0 & 1 \\ 1 & 1 \\ 1 & 0 \end{bmatrix}_{3\times 2} \end{array}$$

EXAMPLE:

For the relation matrix.

$$M = \begin{array}{c} \\ 1 \\ 2 \\ 3 \end{array} \begin{array}{c} \begin{array}{ccc} 1 & 2 & 3 \end{array} \\ \begin{bmatrix} 1 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 1 \end{bmatrix} \end{array}$$

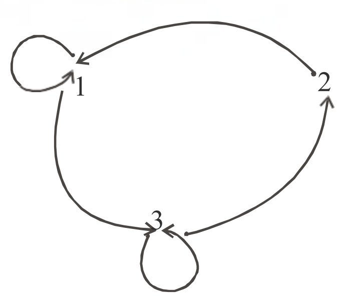
1. List the set of ordered pairs represented by M.
2. Draw the directed graph of the relation.

SOLUTION:

The relation corresponding to the given Matrix is

- R = {(1,1), (1,3), (2,1), (3,2), (3,3)}

And its Directed graph is given below



EXERCISE:

Let $A = \{2, 4\}$ and $B = \{6, 8, 10\}$ and define relations R and S from A to B as follows:

for all $(x,y) \in A \times B$, $x \, R \, y \Leftrightarrow x \mid y$

for all $(x,y) \in A \times B$, $x \, S \, y \Leftrightarrow y - 4 = x$

State explicitly which ordered pairs are in $A \times B$, R, S, $R \cup S$ and $R \cap S$.

SOLUTION

$A \times B = \{(2,6), (2,8), (2,10), (4,6), (4,8), (4,10)\}$

$R = \{(2,6), (2,8), (2,10), (4,8)\}$

$S = \{(2,6), (4,8)\}$

$R \cup S = \{(2,6), (2,8), (2,10), (4,8)\} = R$

$R \cap S = \{(2,6), (4,8)\} = S$

Types of Relations

REFLEXIVE RELATION:

Let R be a relation on a set A. R is reflexive if and only if, for all $a \in A$, $(a, a) \in R$ or equivalently aRa. That is, each element of A is related to itself.

REMARK

R is not reflexive iff there is an element "a" in A such that $(a, a) \notin R$. That is, some element "a" of A is not related to itself.

EXAMPLE:

Let A = {1, 2, 3, 4} and define relations R_1, R_2, R_3, R_4 on A as follows:

$R_1 = \{(1, 1), (3, 3), (2, 2), (4, 4)\}$
$R_2 = \{(1, 1), (1, 4), (2, 2), (3, 3), (4, 3)\}$
$R_3 = \{(1, 1), (1, 2), (2, 1), (2, 2), (3, 3), (4, 4)\}$
$R_4 = \{(1, 3), (2, 2), (2, 4), (3, 1), (4, 4)\}$

Then,

R_1 is reflexive, since $(a, a) \in R_1$ for all $a \in A$.
R_2 is not reflexive, because $(4, 4) \notin R_2$.
R_3 is reflexive, since $(a, a) \in R_3$ for all $a \in A$.
R_4 is not reflexive, because $(1, 1) \notin R_4$, $(3, 3) \notin R_4$

DIRECTED GRAPH OF A REFLEXIVE RELATION:

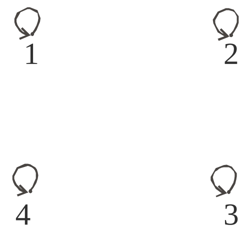
The directed graph of every reflexive relation includes an arrow from every point to the point itself (i.e., a loop).

EXAMPLE :

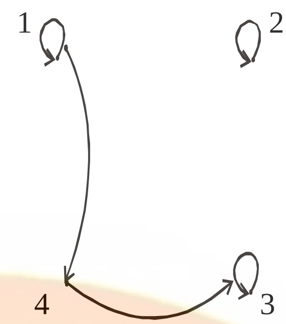
Let A = {1, 2, 3, 4} and define relations R_1, R_2, R_3, and R_4 on A by

$R_1 = \{(1, 1), (3, 3), (2, 2), (4, 4)\}$
$R_2 = \{(1, 1), (1, 4), (2, 2), (3, 3), (4, 3)\}$
$R_3 = \{(1, 1), (1, 2), (2, 1), (2, 2), (3, 3), (4, 4)\}$
$R_4 = \{(1, 3), (2, 2), (2, 4), (3, 1), (4, 4)\}$

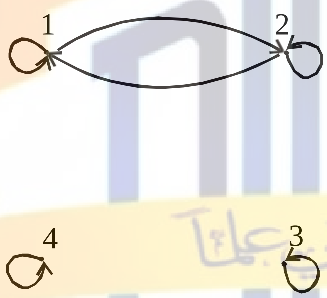
Then their directed graphs are the following:



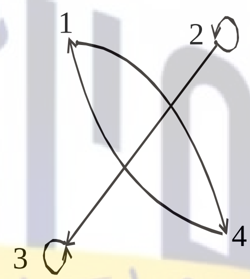
R_1 is reflexive because at every point of the set A we have a loop in the graph.



R_2 is not reflexive, as there is no loop at 4.



R_3 is reflexive



R_4 is not reflexive, as there are no loops at 1and 3.

<u>MATRIX REPRESENTATION OF A REFLEXIVE RELATION:</u>

Let A = $\{a_1, a_2, \ldots, a_n\}$. A Relation R on A is reflexive if and only if $(a_i, a_j) \in R \ \forall$ i=1,2, …,n.

Accordingly, R is **reflexive** if all the elements on the **main diagonal** of the matrix **M** representing R are equal to 1.

EXAMPLE:

The relation R = {(1,1), (1,3), (2,2), (3,2), (3,3)} on A = {1,2,3} represented by the following matrix M, is reflexive.

$$M = \begin{array}{c} \\ 1 \\ 2 \\ 3 \end{array}\begin{array}{c} \begin{array}{ccc} 1 & 2 & 3 \end{array} \\ \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix} \end{array}$$

SYMMETRIC RELATION

Let R be a relation on a set A. R is symmetric if, and only if, for all a, b $\in$ A, if (a, b)$\in$R, then (b, a) $\in$R.

That is, if aRb then bRa.

<u>REMARK</u>

R is not symmetric iff there are elements *a* and *b* in A such that (a, b) $\in$R, but (b, a) $\notin$R.

EXAMPLE

Let $A = \{1, 2, 3, 4\}$ and define relations R_1, R_2, R_3, and R_4 on A as follows.

$R_1 = \{(1, 1), (1, 3), (2, 4), (3, 1), (4,2)\}$
$R_2 = \{(1, 1), (2, 2), (3, 3), (4, 4)\}$
$R_3 = \{(2, 2), (2, 3), (3, 4)\}$
$R_4 = \{(1, 1), (2, 2), (3, 3), (4, 3), (4, 4)\}$

Then R_1 is symmetric because for every order pair (a, b) in R_1 also have (b, a) in R_1. For example, we have (1, 3) in R_1 then we have (3, 1) in R_1. Similarly all other ordered pairs can be checked.

R_2 is also symmetric. We say it is vacuously true.

R_3 is not symmetric, because $(2,3) \in R_3$ but $(3,2) \notin R_3$.

R_4 is not symmetric because $(4,3) \in R_4$ but $(3,4) \notin R_4$.

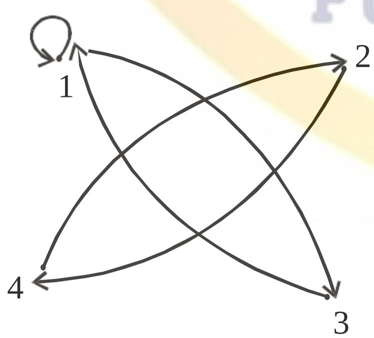
DIRECTED GRAPH OF A SYMMETRIC RELATION

For a symmetric directed graph whenever there is an arrow going from one point of the graph to a second, there is an arrow going from the second point back to the first.

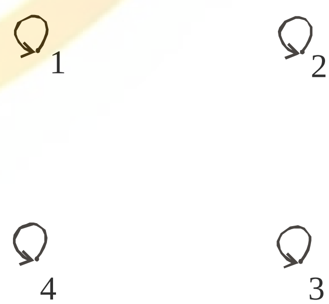
EXAMPLE

Let $A = \{1, 2, 3, 4\}$ and define relations R_1, R_2, R_3 and R_4 on A by the directed graphs:

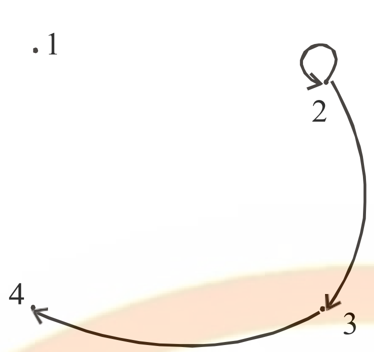
$R_1 = \{(1, 1), (1, 3), (2, 4), (3, 1), (4,2)\}$
$R_2 = \{(1, 1), (2, 2), (3, 3), (4, 4)\}$
$R_3 = \{(2, 2), (2, 3), (3, 4)\}$
$R_4 = \{(1, 1), (2, 2), (3, 3), (4, 3), (4, 4)\}$

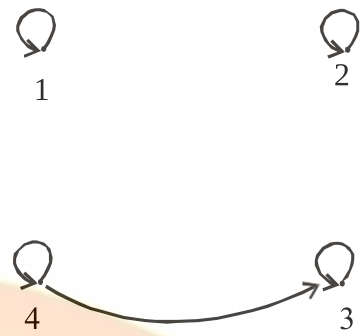


R_1 is symmetric



R_2 is symmetric





R_3 is not symmetric since there are arrows from 2 to 3 and from 3 to 4 but not conversely

R_4 is not symmetric since there is an arrow from 4 to 3 but no arrow from 3 to 4

MATRIX REPRESENTATION OF A SYMMETRIC RELATION

Let

$A = \{a_1, a_2, \ldots, a_n\}$.

The relation R on A is symmetric if and only if for all $a_i, a_j \in A$,

if $(a_i, a_j) \in R$ then $(a_j, a_i) \in R$.

Accordingly, R is symmetric if the elements in the ith row are the same as the elements in the ith column of the matrix M representing R. More precisely, M is a symmetric matrix i.e. $M = M^t$

EXAMPLE:

The relation $R = \{(1,3), (2,2), (3,1), (3,3)\}$ on $A = \{1,2,3\}$ represented by the following matrix M is symmetric.

$$M = \begin{array}{c} \\ 1 \\ 2 \\ 3 \end{array}\begin{array}{c} \begin{array}{ccc} 1 & 2 & 3 \end{array} \\ \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix} \end{array}$$

TRANSITIVE RELATION

Let R be a relation on a set A. R is transitive if and only if for all $a, b, c \in A$,

if $(a, b) \in R$ and $(b, c) \in R$ then $(a, c) \in R$.

That is, if aRb and bRc then aRc.

In words, if any one element is related to a second and that second element is related to a third, then the first is related to the third.

<u>**Note**</u>: The "first", "second" and "third" elements need not to be distinct.

<u>**REMARK**</u>

R is not transitive iff there are elements a, b, c in A such that if (a, b) $\in$R and (b, c) $\in$R but (a, c) $\notin$R.

EXAMPLE

Let A = {1, 2, 3, 4} and define relations R_1, R_2 and R_3 on A as follows:

R_1 = {(1, 1), (1, 2), (1, 3), (2, 3)}

R_2 = {(1, 2), (1, 4), (2, 3), (3, 4)}

R_3 = {(2, 1), (2, 4), (2, 3), (3,4)}

Then R_1 is transitive because (1, 1), (1, 2) are in R, then to be transitive relation (1,2) must be there and it belongs to R.

Similarly for other order pairs. R_2 is not transitive since (1,2) and (2,3) $\in$ R_2 but (1,3) $\notin$ R_2.

R_3 is transitive.

DIRECTED GRAPH OF A TRANSITIVE RELATION

For a transitive directed graph, whenever there is an arrow going from one point to the second, and from the second to the third, there is an arrow going directly from the first to the third.

EXAMPLE

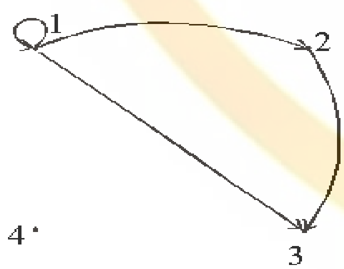
Let A = {1, 2, 3, 4} and define relations R_1, R_2 and R_3 on A by the directed graphs:

R_1 = {(1, 1), (1, 2), (1, 3), (2, 3)}

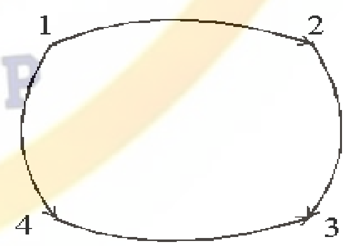
R_2 = {(1, 2), (1, 4), (2, 3), (3, 4)}

R_3 = {(2, 1), (2, 4), (2, 3), (3,4)}

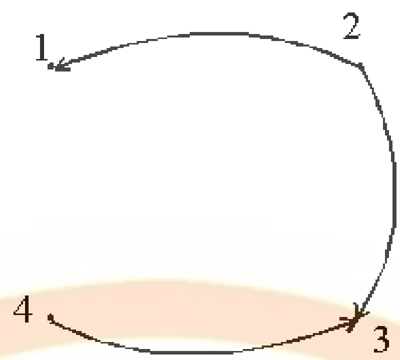
SOLUTION:



R_1 is transitive



R_2 is not transitive since there is an arrow from 1 to 2 and from 2 to 3 but no arrow from 1 to 3 directly



R_3 is transitive

EXERCISE:

Let A = {1, 2, 3, 4} and define the null relation ϕ and universal relation $A \times A$ on A. Test these relations for reflexive, symmetric and transitive properties.

SOLUTION:

Reflexive:

a) $\varnothing$ is not reflexive since $(1,1), (2,2), (3,3), (4,4) \notin \varnothing$.

b) $A \times A$ is reflexive since $(a,a) \in A \times A$ for all $a \in A$.

Symmetric

a) For the null relation $\varnothing$ on A to be symmetric, it must satisfy the implication:

if $(a,b) \in \varnothing$ then $(a, b) \in \varnothing$.

Since $(a, b) \in \varnothing$ is never true, the implication is vacuously true or true by default. Hence $\varnothing$ is symmetric.

b) The universal relation $A \times A$ is symmetric, for it contains all ordered pairs of elements of A. Thus,

if $(a, b) \in A \times A$ then $(b, a) \in A \times A$ for all a, b in A.

Transitive

a) The null relation $\varnothing$ on A is transitive, because the implication.

if $(a, b) \in \varnothing$ and $(b, c) \in \varnothing$ then $(a, c) \in \varnothing$ is true by default,

since the condition $(a, b) \in \varnothing$ is always false.

b) The universal relation $A \times A$ is transitive for it contains all ordered pairs of elements of A.

Accordingly, if $(a, b) \in A \times A$ and $(b, c) \in A \times A$ then $(a, c) \in A \times A$

EXERCISE:

Let A = {0, 1, 2} and

R = {(0,2), (1,1), (2,0)} be a relation on A.

1. Is R reflexive? Symmetric? Transitive?
2. Which ordered pairs are needed in R to make it a reflexive and transitive relation.

SOLUTION:

1. R is not reflexive, since $0 \in A$ but $(0, 0) \notin R$ and also $2 \in A$ but $(2, 2) \notin R$.
   R is clearly symmetric.
   R is not transitive, since $(0, 2)$ & $(2, 0) \in R$ but $(0, 0) \notin R$.

2. For R to be reflexive, it must contain ordered pairs (0,0) and (2,2).
   For R to be transitive,
   we note (0,2) and $(2,0) \in$ but $(0,0) \notin R$.
   Also (2,0) and $(0,2) \in R$ but $(2,2) \notin R$.
   Hence (0,0) and (2,2). Are needed in R to make it a transitive relation.

EXERCISE:

Define a relation L on the set of real numbers **R** be defined as follows:

for all $x, y \in R$, $\quad x\ L\ y \Leftrightarrow x < y$.

a. Is L reflexive?
b. Is L symmetric?
c. Is L transitive?

SOLUTION:

a. L is not reflexive, because $x \underline{<} x$ for any real number x. (e.g. $1 \nless 1$)

b. L is not symmetric, because for all $x, y \in R$, if $x < y$ then $y \nless x$
(e.g. $0 < 1$ but $1 \nless 0$)

c. L is transitive, because for all, $x, y, z \in R$, if $x < y$ and $y < z$, then $x < z$.
(by transitive law of order of real numbers).

EXERCISE:

Define a relation R on the set of positive integers Z^+ as follows:

for all $a, b \in Z^+$, a R b *iff* $a \times b$ is odd.

Determine whether the relation is

a. reflexive b. symmetric c. transitive

SOLUTION:

Firstly, recall that the product of two positive integers is odd if and only if both of them are odd.

a. reflexive
R is not reflexive, because $2 \in Z^+$ but 2 R 2
for $2 \times 2 = 4$ which is not odd.

b. symmetric
R is symmetric, because
if a R b then $a \times b$ is odd or equivalently $b \times a$ is odd
$(b \times a = a \times b) \Rightarrow$ b R a.

c. transitive
R is transitive, because if a R b then $a \times b$ is odd
$\Rightarrow$ both "a" and "b" are odd. Also bRc means $b \times c$ is odd
$\Rightarrow$ both "b" and "c" are odd.

Now if aRb and bRc, then all of a, b, c are odd and so $a \times c$ is odd. Consequently aRc.

EXERCISE:

Let "D" be the "divides" relation on Z defined as:

for all m, n $\in$Z, m D n$\Leftrightarrow$ m|n

Determine whether D is reflexive, symmetric or transitive. Justify your answer.

SOLUTION:

Reflexive

Let m $\in$Z, since every integer divides itself.

So m|m $\forall$ m $\in$Z therefore mDm $\forall$ m $\in$Z

Accordingly D is reflexive

Symmetric

Let m, n $\in$ Z and suppose m D n.

By definition of D, this means m|n (i.e.= an integer)

Clearly, then it is not necessary that = an integer.

Accordingly, if m D n then n D m, $\forall$ m, n $\in$Z

Hence D is not symmetric.

Transitive

Let m, n, p $\in$Z and suppose m D n and n D p.

Now m D n $\Rightarrow$ m|n $\Rightarrow$ = an integer.

Also n D p $\Rightarrow$ n|p $\Rightarrow$ = an integer.

We note $\frac{p}{m} = \left(\frac{p}{n}\right) * \left(\frac{n}{m}\right)$ = (an int) * (an int)

= an int

$\Rightarrow$ m|p and so mDp

Thus if mDn and nDp then mDp $\forall$ m, n, p $\in$Z

Hence D is transitive.

EXERCISE:

Let A be the set of people living in the world today. A binary relation R is defined on A as follows:

for all p, q $\in$A, pRq $\Leftrightarrow$ p has the same first name as q.

Determine whether the relation R is reflexive, symmetric and/or transitive.

SOLUTION:

Reflexive

Since every person has the same first name as his/her self.

Hence for all p $\in$ A, pRp. Thus, R is reflexive.

Symmetric:

Let p, q $\in$A and suppose pRq.

p has the same first name as q.

q has the same first name as p.

q R p

Thus if pRq then qRp $\forall$ p,q $\in$A.

R is symmetric.

Transitive

Let p, q, s $\in$A and suppose p R q and qRr.

Now $pRq \Leftrightarrow p$ has the same first name as q
and $qRr \Leftrightarrow q$ has the same first name as r.
Consequently, p has the same first name as r.
$\Rightarrow$ pRr
Thus, if pRq and qRs then pRr, $\forall$ p, q, r $\in$ A.
Hence R is transitive.

EQUIVALENCE RELATION:

Let A be a non-empty set and R a binary relation on A. R is an equivalence relation if, and only if, R is reflexive, symmetric, and transitive.

EXAMPLE:

Let A = {1, 2, 3, 4} and
R = {(1,1), (2,2), (2,4), (3,3), (4,2), (4,4)}
be a binary relation on A.

Note that R is reflexive, symmetric and transitive, hence an equivalence relation.

CONGRUENCES:

Let m and n be integers and d be a positive integer. The notation

$m \equiv n \pmod{d}$ means that

$d \mid (m - n)$ {d divides m minus n}.There exists an integer k such that

$(m - n) = d \cdot k$

EXAMPLE:

c. Is $22 \equiv 1 \pmod{3}$?
b. Is $-5 \equiv +10 \pmod{3}$?
d. Is $7 \equiv 7 \pmod{3}$?
d. Is $14 \equiv 4 \pmod{3}$?

SOLUTION

a. Since $22-1 = 21 = 3\times7$.
Hence $3|(22-1)$, and so $22 \equiv 1 \pmod{3}$
b. Since $-5 - 10 = -15 = 3 \times (-5)$,
Hence $3|((-5)-10)$, and so $-5 \equiv 10 \pmod{3}$
c. Since $7 - 7 = 0 = 3 \times 0$
Hence $3|(7-7)$, and so $7 \equiv 7 \pmod{3}$
d. Since $14 - 4 = 10$, and $3 / 10$ because $10 \neq 3\cdot k$ for any integer k. Hence $14 \equiv 4 \pmod{3}$.

EXERCISE:

Define a relation R on the set of all integers Z as follows:

for all integers m and n, $m\,R\,n \Leftrightarrow m \equiv n \pmod{3}$

Prove that R is an equivalence relation.

SOLUTION:

R is reflexive.

R is reflexive iff for all $m \in Z$, m R m.
By definition of R, this means that
For all $m \in Z$, $m \equiv m \pmod{3}$
Since $m - m = 0 = 3 \times 0$.
Hence $3|(m-m)$, and so $m \equiv m \pmod{3}$
mRm
$\Rightarrow$ R is reflexive.

R is symmetric.

R is symmetric iff for all m, n $\in Z$
if m R n then n R m.
Now mRn
$\Rightarrow$ $m \equiv n \pmod 3$
$\Rightarrow$ $3|(m-n)$
$\Rightarrow$ $m-n = 3k$, for some integer k.
$\Rightarrow$ $n - m = 3(-k)$, $-k \in Z$
$\Rightarrow$ $3|(n-m)$
$\Rightarrow$ $n \equiv m \pmod 3$
$\Rightarrow$ nRm

Hence R is symmetric.

R is transitive

R is transitive iff for all m, n, p $\in Z$,
if mRn and nRp then mRp
Now mRn and nRp means $m \equiv n \pmod 3$ and $n \equiv p \pmod 3$
$3|(m-n)$ and $3|(n-p)$
$(m-n) = 3r$ and $(n-p) = 3s$ for some r, s $\in Z$
Adding these two equations, we get,
$(m - n) + (n - p) = 3r + 3s$
$m - p = 3(r + s)$, where $r + s \in Z$
$3|(m - p)$
$m \equiv p \pmod 3 \Leftrightarrow m\,R\,p$

Hence R is transitive. R being reflexive, symmetric and transitive, is an equivalence relation.

Matrix Representation of Relations

EXERCISE:

Suppose R and S are binary relations on a set A.

a. If R and S are reflexive, is $R \cap S$ reflexive?
b. If R and S are symmetric, is $R \cap S$ symmetric?
c. If R and S are transitive, is $R \cap S$ transitive?

SOLUTION:

a. $R \cap S$ is reflexive:

Suppose R and S are reflexive.

Then by definition of reflexive relation

$\forall\ a \in A\ (a,a) \in R$ and $(a,a) \in S$

$\Rightarrow\ \forall\ a \in A\ (a,a) \in R \cap S$

(by definition of intersection)

Accordingly, $R \cap S$ is reflexive.

b. $R \cap S$ is symmetric.

Suppose R and S are symmetric.

To prove $R \cap S$ is symmetric we need to show that

$\forall\ a, b \in A$, if $(a,b) \in R \cap S$ then $(b,a) \in R \cap S$.

Suppose $(a,b) \in R \cap S$.

$\Rightarrow\ (a,b) \in R$ and $(a,b) \in S$

(by the definition of Intersection of two sets)

Since R is symmetric, therefore if $(a,b) \in R$ then

$(b,a) \in R$. Similarly S is symmetric, so if $(a,b) \in S$ then $(b,a) \in S$.

Thus $(b,a) \in R$ and $(b,a) \in S$

$\Rightarrow\ (b,a) \in R \cap S$ (by definition of intersection)

Accordingly, $R \cap S$ is symmetric.

c. $R \cap S$ is transitive.

Suppose R and S are transitive.

To prove $R \cap S$ is transitive we must show that

$\forall\ a,b,c, \in A$, if $(a,b) \in R \cap S$ and $(b,c) \in R \cap S$

then $(a,c) \in R \cap S$.

Suppose $(a,b) \in R \cap S$ and $(b,c) \in R \cap S$

$\Rightarrow\ (a,b) \in R$ and $(a,b) \in S$ and $(b,c) \in R$ and $(b,c) \in S$

Since R is transitive, therefore

if $(a,b) \in R$ and $(b,c) \in R$ then $(a,c) \in R$.

Also S is transitive, so $(a,c) \in S$

Hence we conclude that $(a,c) \in R$ and $(a,c) \in S$

and so $(a,c) \in R \cap S$ (by definition of intersection)

Accordingly, $R \cap S$ is transitive.

EXAMPLE:

Let $A = \{1,2,3,4\}$

and let R and S be transitive binary relations on A defined as:

$R = \{(1,2), (1,3), (2,2), (3,3), (4,2), (4,3)\}$

and $S = \{(2,1), (2,4), (3,3)\}$

Then $R \cup S = \{(1,2), (1,3), (2,1), (2,2), (2,4), (3,3), (4,2), (4,3)\}$

We note $(1,2)$ and $(2,1) \in R \cup S$, but $(1,1) \notin R \cup S$

Hence $R \cup S$ is not transitive.

IRREFLEXIVE RELATION:

Let R be a binary relation on a set A. R is irreflexive iff for all $a \in A, (a,a) \notin R$.

That is, R is irreflexive if no element in A is related to itself by R.

REMARK:

R is not irreflexive iff there is an element $a \in A$ such that $(a,a) \in R$.

EXAMPLE:

Let $A = \{1,2,3,4\}$ and define the following relations on A:

$R_1 = \{(1,3), (1,4), (2,3), (2,4), (3,1), (3,4)\}$

$R_2 = \{(1,1), (1,2), (2,1), (2,2), (3,3), (4,4)\}$

$R_3 = \{(1,2), (2,3), (3,3), (3,4)\}$

Then R_1 is irreflexive since no element of A is related to itself in R_1. i.e.
$(1,1) \notin R_1$, $(2,2) \notin R_1$, $(3,3) \notin R_1$, $(4,4) \notin R_1$

R_2 is not irreflexive, since all elements of A are related to themselves in R_2

R_3 is not irreflexive since $(3,3) \in R_3$. Note that R_3 is not reflexive.

NOTE:

A relation may be neither reflexive nor irreflexive.

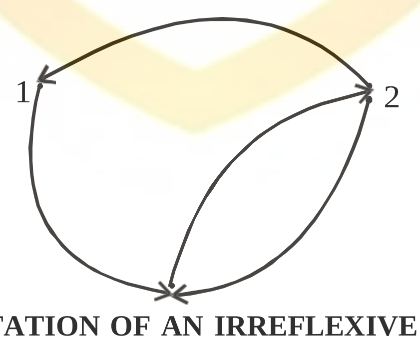
DIRECTED GRAPH OF AN IRREFLEXIVE RELATION:

Let R be an **irreflexive** relation on a set A. Then by definition, no element of A is related to itself by R. Accordingly, there is no loop at each point of A in the directed graph of R.

EXAMPLE:

Let $A = \{1,2,3\}$

and $R = \{(1,3), (2,1), (2,3), (3,2)\}$ be represented by the directed graph.



MATRIX REPRESENTATION OF AN IRREFLEXIVE RELATION

Let R be an irreflexive relation on a set A. Then by definition, no element of A is related to itself by R.

Since the self related elements are represented by 1's on the main diagonal of the matrix representation of the relation, so for irreflexive relation R, the matrix will contain all 0's in its main diagonal.

It means that a relation is irreflexive if in its matrix representation the diagonal elements are all zero, if one of them is not zero the we will say that the relation is not irreflexive.

EXAMPLE:

Let A = {1,2,3} and R = {(1,3), (2,1), (2,3), (3,2)} be represented by the matrix

$$M = \begin{array}{c} \\ 1 \\ 2 \\ 3 \end{array} \begin{array}{c} \begin{array}{ccc} 1 & 2 & 3 \end{array} \\ \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} \end{array}$$

Then R is irreflexive, since all elements in the main diagonal are 0's.

EXERCISE:

Let R be the relation on the set of integers Z defined as:
for all $a,b \in Z$, $(a,b) \in R \Leftrightarrow a > b$.
Is R irreflexive?

SOLUTION:

R is irreflexive if for all $a \in Z$, $(a,a) \notin R$.
Now by the definition of given relation R,
for all $a \in Z$, $(a,a) \notin R$ since $a \not> a$.
Hence R is irreflexive.

ANTISYMMETRIC RELATION:

Let R be a binary relation on a set A.R is a**nti-symmetric** iff
$\forall a, b \in A$ if $(a,b) \in R$ and $(b,a) \in R$ then $a = b$.

REMARK:

1) R is not **anti-symmetric** iff there are elements a and b in A such that $(a, b) \in R$ and $(b, a) \in R$ but $a \neq b$.
2) The properties of being **symmetric** and being **anti-symmetric** are not negative of each other.

EXAMPLE:

Let A = {1,2,3,4} and define the following relations on A.

R_1 = {(1,1),(2,2),(3,3)}
R_2 = {(1,2),(2,2), (2,3), (3,4), (4,1)}
R_3={(1,3),(2,2), (2,4), (3,1), (4,2)}
R_4={(1,3),(2,4), (3,1), (4,3)}

R_1 is anti-symmetric and symmetric .

R_2 is anti-symmetric but not symmetric because $(1,2) \in R_2$ but $(2,1) \notin R_2$.

R_3 is not anti-symmetric since (1,3) & $(3,1) \in R_3$ but $1 \neq 3$.

Note that R_3 is symmetric.

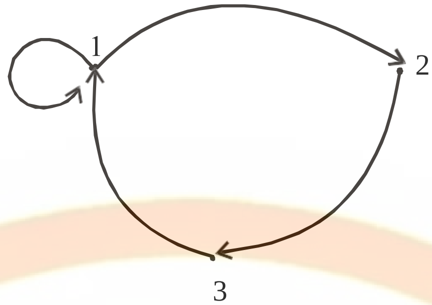
R_4 is neither anti-symmetric because (1,3) & $(3,1) \in R_4$ but $1 \neq 3$ nor symmetric because $(2,4) \in R4$ but $(4,2) \notin R4$

DIRECTED GRAPH OF AN ANTISYMMETRIC RELATION:

Let R be an anti-symmetric relation on a set A. Then by definition, no two distinct elements of A are related to each other. Accordingly, there is no pair of arrows between two distinct elements of A in the directed graph of R.

EXAMPLE:

Let A = {1,2,3} And R be the relation defined on A is R ={(1,1), (1,2), (2,3), (3,1)}.Thus R is represented by the directed graph as



R is anti-symmetric, since there is no pair of arrows between two distinct points in A.

MATRIX REPRESENTATION OF AN ANTISYMMETRIC RELATION:

Let R be an anti-symmetric relation on a set A = $\{a_1, a_2, \ldots, a_n\}$. Then if $(a_i, a_j) \in R$ for $i \neq j$ then $(a_i, a_j) \notin R$.
Thus in the matrix representation of R there is a 1 in the ith row and jth column iff the jth row and ith column contains 0 vice versa.

EXAMPLE:

Let A = {1,2,3} and a relation R = {(1,1), (1,2), (2,3), (3,1)}on A be represented by the matrix.

$$M = \begin{array}{c} \\ 1 \\ 2 \\ 3 \end{array} \begin{array}{c} \begin{array}{ccc} 1 & 2 & 3 \end{array} \\ \begin{bmatrix} 1 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix} \end{array}$$

Then R is anti-symmetric as clear by the form of matrix M

PARTIAL ORDER RELATION:

Let R be a binary relation defined on a set A. R is a partial order relation,if and only if, R is **reflexive, antisymmetric**, and **transitive**. The set A together with a partial ordering R is called a **partially ordered** set or **poset.**

EXAMPLE:

Let R be the set of real numbers and define the“less than or equal to” , on R as follows:

for all real numbers x and y in R.$x \leq y \Leftrightarrow x < y$ or $x = y$

Show that ≤ is a partial order relation.

SOLUTION:

≤ is reflexive

For ≤ to be reflexive means that $x \leq x$ for all $x \in R$

But $x \leq x$ means that $x < x$ or $x = x$ and $x = x$ is always true.
Hence under this relation every element is related to itself.

≤ is anti-symmetric.

For $\leq$ to be anti-symmetric means that

$\forall$ x, y $\in$R, if x $\leq$ y and y $\leq$ x, then x = y.

This follows from the definition of $\leq$ and the trichotomy property, which says that

"given any real numbers x and y, exactly one of the following holds:

x < y or x = y or x > y"

$\leq$ is transitive

For $\leq$ to be transitive means that

$\forall$ x, y, z $\in$R, if x$\leq$ y and y $\leq$ z then x $\leq$ z.

This follows from the definition of $\leq$ and the transitive property of order of real numbers, which says that "given any real numbers x, y and z,

if x < y and y < z then x < z"

Thus $\leq$ being reflexive, anti-symmetric and transitive is a partial order relation on R.

EXERCISE:

Let A be a non-empty set and P(A) the power set of A.

Define the "subset" relation, $\subseteq$, as follows:

for all X,Y $\in$ P(A), X $\subseteq$ Y $\Leftrightarrow$ $\forall$ x, iff x $\in$X then x $\in$Y.

Show that $\subseteq$ is a partial order relation.

SOLUTION:

1. **$\subseteq$ is reflexive**

Let X $\in$ P(A). Since every set is a subset of itself, therefore

X $\subseteq$ X, $\forall$ X $\in$P(A).

Accordingly $\subseteq$ is reflexive.

2. **$\subseteq$ is anti-symmetric**

Let X, Y $\in$P(A) and suppose X $\subseteq$ Y and Y $\subseteq$ X.Then by definition of equality of two sets it follows that X = Y.

Accordingly, $\subseteq$ is anti-symmetric.

3. **$\subseteq$ is transitive**

Let X, Y, Z $\in$P(A) and suppose X $\subseteq$ Y and Y $\subseteq$ Z. Then by the transitive property of subsets "if U $\subseteq$ V and V $\subseteq$ W then U $\subseteq$ W"it follows X $\subseteq$ Z.

Accordingly $\subseteq$ is transitive.

EXERCISE:

Let "|" be the "divides" relation on a set A of positive integers.

That is, for all a, b $\in$A, a|b $\Leftrightarrow$ b = k ·a for some integer k.

Prove that | is a partial order relation on A.

SOLUTION:

1. **"|" is reflexive.** [We must show that, $\forall$ a $\in$A, a|a]

Suppose a $\in$A. Then a = 1·a and so a|a by definition of divisibility.

2. **"|" is anti-symmetric**

[We must show that for all a, b $\in$A, if a|b and b|a then a=b]

Suppose a|b and b|a

By definition of divides there are integers k_1, and k_2 such that

$$b = k_1 \cdot a \qquad \text{and} \qquad a = k_2 \cdot b$$

Now $\quad b = k_1 \cdot a$

$\qquad = k_1 \cdot (k_2 \cdot b)$ (by substitution)

$\qquad = (k_1 \cdot k_2) \cdot b$

Dividing both sides by b gives

$$1 = k_1 \cdot k_2$$

Since a, b $\in$ A, where A is the set of positive integers, so the equations

$$b = k_1 \cdot a \qquad \text{and} \qquad a = k_2 \cdot b$$

implies that k_1 and k_2 are both positive integers. Now the equation

$$k_1 \cdot k_2 = 1$$

can hold only when $k_1 = k_2 = 1$

Thus $a = k_2 \cdot b = 1 \cdot b = b$ i.e., a = b

3. "|" is transitive

[We must show that $\forall a,b,c \in A$ if a|b and b|c than a|c]

Suppose a|b and b|c

By definition of divides, there are integers k_1 and k_2 such that

$$b = k_1 \cdot a \qquad \text{and} \qquad c = k_2 \cdot b$$

Now $\quad c = k_2 \cdot b$

$\qquad = k_2 \cdot (k_1 \cdot a)$ (by substitution)

$\qquad = (k_2 \cdot k_1) \cdot a$ (by associative law under multiplication)

$\qquad = k_3 \cdot a$ where $k_3 = k_2 \cdot k_1$ is an integer

$\qquad \Rightarrow$ a|c by definition of divides

Thus "|" is a partial order relation on A.

EXERCISE:

Let "R" be the relation defined on the set of integers Z as follows:

for all a, b $\in$ Z, aRb iff $b = a^r$ for some positive integer r.

Show that R is a partial order on Z.

SOLUTION:

1. **R is Reflexive**

   Let a $\in$ Z, then $a = a^r$ for r = 1, so aRa.

   So R is reflexive.

2. **R is ant-symmetric.**

Let a, b $\in$ Z and suppose aRb and bRa. Then there are positive integers r and s such that

$$b = a^r \qquad \text{and} \qquad a = b^s$$

Now, $\quad a = b^s$

$\qquad = (a^r)^s$ by substitution

$\qquad = a^{rs}$

$$\Rightarrow \quad rs = 1$$

Since r and s are positive integers, so this equation can hold if, and only if, r =1

and s = 1 and then $a = b^s = b^1 = b$ i.e., a = b
Thus R is anti-symmetric.

3. R is transitive.

Let a, b, c ∈Z and suppose aRb and bRc**.**
Then there are positive integers r and s such that

$$b = a^r \quad \text{and} \quad c = b^s$$

Now $c = b^r$

$= (a^r)^s$ (by substitution)

$= a^{rs} = a^t$ (where t = rs is also a positive integer)

Hence by definition of R, aRc. Therefore, R is transitive.
Accordingly, R is a **partial order** relation on Z.

Inverse of Relations

INVERSE OF A RELATION:

Let R be a relation from A to B. The inverse relation R^{-1} from B to A is defined as:

$$R^{-1} = \{(b,a) \in B \times A \mid (a,b) \in R\}$$

More simply, the inverse relation R^{-1} of R is obtained by interchanging the elements of all the ordered pairs in R.

EXAMPLE:

Let A = {2, 3, 4} and B = {2,6,8}and let R be the "divides" relation from A to B i.e. for all $(a,b) \in A \times B$, $a\ R\ b \Leftrightarrow a \mid b$ (a divides b)

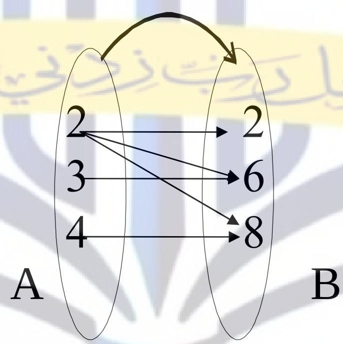
Then R = {(2,2), (2,6), (2,8), (3,6), (4,8)}and R^{-1}= {(2,2), (6,2), (8,2), (6,3), (8,4)}

In words, R^{-1} may be defined as:

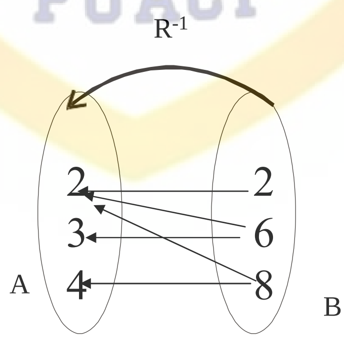
for all $(b,a) \in B \times A$, $b\ R\ a \Leftrightarrow b$ is a multiple of a.

ARROW DIAGRAM OF AN INVERSE RELATION:

The relation R = {(2,2), (2,6), (2,8), (3,6), (4,8)} is represented by the arrow diagram.



Then inverse of the above relation can be obtained simply changing the directions of the arrows and hence the diagram is



MATRIX REPRESENTATION OF INVERSE RELATION:

The relation R = {(2, 2), (2, 6), (2, 8), (3, 6), (4, 8)}from A = {2, 3,4} to B = {2, 6, 8} is defined by the matrix M below:

$$M = \begin{array}{c} \\ 2 \\ 3 \\ 4 \end{array}\begin{array}{c} \begin{array}{ccc} 2 & 6 & 8 \end{array} \\ \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \end{array} \qquad M^t = \begin{array}{c} \\ 2 \\ 6 \\ 8 \end{array}\begin{array}{c} \begin{array}{ccc} 2 & 3 & 4 \end{array} \\ \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix} \end{array}$$

The matrix representation of inverse relation R^{-1} is obtained by simply taking its transpose. (i.e., changing rows by columns and columns by rows). Hence R-1 is represented by Mt as shown.

EXERCISE:

Let R be a binary relation on a set A. Prove that:

(i) If R is reflexive, then R^{-1} is reflexive.
(ii) If R is symmetric, then R^{-1} is symmetric.
(iii) If R is transitive, then R^{-1} is transitive.
(iv) If R is antisymmetric, then R^{-1} is antisymmetric.

SOLUTION (i) **If R is reflexive, then R^{-1} is reflexive.**

Suppose that the relation R on A is reflexive. By definition, $\forall$ a $\in$ A, (a, a) $\in$R. Since $\mathbf{R^{-1}}$ consists of exactly those ordered pairs which are obtained by interchanging the first and second element of ordered pairs in R, therefore, if (a, a) $\in$ R then (a, a) $\in$ R^{-1}. Accordingly, $\forall$ a $\in$ A, (a, a) $\in$ R^{-1}.
Hence $R^{-1}$is reflexive as well.

SOLUTION (ii) Suppose that the relation R on A is symmetric.

Let (a, b) $\in$ $R^{-1}$for a,b $\in$A. By definition of R^{-1}, (b, a) $\in$R. Since R is symmetric, therefore (a, b) $\in$R. But then by definition of R^{-1}, (b, a) $\in$R.
We have thus shown that for all a, b $\in$ A, if (a, b) $\in$R-1 then (b, a) $\in$ R^{-1}.
Accordingly $R^{-1}$is symmetric.

SOLUTION (iii) **Prove that if R is transitive, then R-1 is transitive.**

Suppose that the relation R on A is transitive. Let (a, b) $\in$ $R^{-1}$and (b, c) $\in$ R^{-1}.
Then by definition of R^{-1}, (b, a) $\in$R and (c, b) $\in$R. Now R is transitive, therefore if (c, b) $\in$R and (b, a) $\in$R then (c, a) $\in$R.
Again by definition of R^{-1}, we have (a, c) $\in$ R^{-1}. We have thus shown that for all a, b, c $\in$ A, if (a, b) $\in$ $R^{-1}$and (b, c) $\in$ R^{-1}then (a, c) $\in$ R^{-1}.
Accordingly R^{-1} is transitive.

SOLUTION (iv) Prove that if R is anti-symmetric. Then R^{-1} **is anti-symmetric**.

Suppose that relation R on A is anti-symmetric. Let (a,b)$\in$ R^{-1} 1 and (b,a)$\in$ R^{-1}
Then by definition of R^{-1}(b,a) $\in$R and (a,b) $\in$R. Since R is antisymmetric, so if (a,b)$\in$R and (b,a)$\in$R then a = b.Thus we have shown that if (a,b) $\in$ R^{-1} and (b,a) $\in$ R^{-1} then a=b.
Accordingly R^{-1} is antisymmetric.

EXERCISE:

Show that the relation R on a set A is symmetric if, and only if,

$R = R^{-1}$.

SOLUTION:

Suppose the relation R on A is symmetric.

Let $(a,b) \in R$. Since R is symmetric, so $(b,a) \in R$. But by definition of R^{-1} if $(b,a) \in R$ then $(a,b) \in R^{-1}$. Since (a,b) is an arbitrary element of R, so

$R \subseteq R^{-1}$ …………… (1)

Next, let $(c,d) \in R^{-1}$. By definition of R^{-1} $(d,c) \in R$. Since R is symmetric, so $(c,d) \in R$. Thus we have shown that if $(c,d) \in R^{-1}$ then $(c,d) \in R$.Hence

$R^{-1} \subseteq R$…………… (2)

By (1) and (2) it follows that $R = R^{-1}$.

Conversely

suppose $R = R^{-1}$.

We have to show that R is symmetric. Let $(a,b) \in R$.

Now by definition of R^{-1} $(b,a) \in R^{-1}$ .Since $R = R^{-1}$, so $(b,a) \in R^{-1} = R$

Thus we have shown that if $(a,b) \in R$ then $(b,a) \in R$

Accordingly R is symmetric.

COMPLEMENTRY RELATION:

Let R be a relation from a set A to a set B. The complementry relation $\overline{R}$ of R is the set of all those ordered pairs in A×B that do not belong to R.

Symbolically:

$\overline{R} = A \times B - R = \{(a,b) \in A \times B \mid (a,b) \notin R\}$

EXAMPLE:

Let A = {1,2,3} and

R = {(1,1), (1,3), (2,2), (2,3), (3,1)} be a relation on A

Then $\overline{R} = \{(1,2), (2,1), (3,2), (3,3)\}$

EXERCISE:

Let R be the relation $R = \{(a,b) \mid a<b\}$ on the set of integers. Find

a) $\overline{R}$ b) R^{-1}

SOLUTION:

a) $\overline{R} = Z \times Z - R = \{(a,b) \mid a \nless b\}$

$= \{(a,b) \mid a \geq b\}$

b) $R^{-1} = \{(a,b) \mid a > b\}$

EXERCISE:

Let R be a relation on a set A. Prove that R is reflexive iff R is irreflexive

SOLUTION:

Suppose R is reflexive. Then by definition, for all $a \in A$, $(a,a) \in R$

But then by definition of the complementry relation $(a,a) \notin R$, $\forall\, a \in A$.

Accordingly R is irreflexive.

Conversely

if R is irreflexive, then $(a,a) \notin R$, $\forall\, a \in A$.

Hence by definition of R, it follows that $(a,a) \in R$, $\forall\, a \in A$

Accordingly R is reflexive.

EXERCISE:

Suppose that R is a symmetric relation on a set A. Is R also symmetric.

SOLUTION:

Let $(a,b)\in \bar{R}$. Then by definition of $\bar{R}$, $(a,b) \notin R$. Since R is symmetric, so if $(a,b) \notin R$ then $(b,a) \notin R$.

{for $(b,a) \in R$ and $(a,b) \notin R$ will contradict the symmetry property of R}

Now $(b,a) \notin R \Rightarrow (b,a) \in \bar{R}$. Hence if $(a,b) \in \bar{R}$ then $(b,a) \in \bar{R}$

Thus $\bar{R}$ is also symmetric.

COMPOSITE RELATION:

Let R be a relation from a set A to a set B and S a relation from B to a set C. The composite of R and S denoted SoR is the relation from A to C, consisting of ordered pairs (a,c) where $a \in A$, $c \in C$, and for which there exists an element $b \in B$ such that $(a,b) \in R$ and $(b,c) \in S$.

Symbolically:

$$SoR = \{(a,c)|a \in A, c \in C, \exists\ b \in B, (a,b) \in R \text{ and } (b,c) \in S\}$$

EXAMPLE:

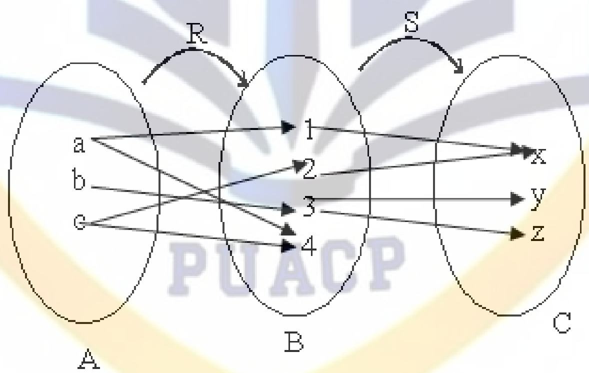
Define R = {(a,1), (a,4), (b,3),(c,1), (c,4)} as a relation from A to B and S = {(1,x),(2,x), (3,y), (3,z)} be a relation from B to C.

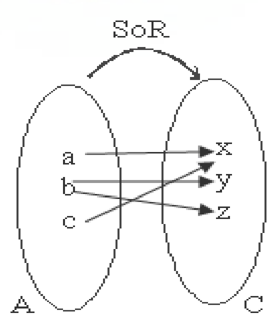
Hence

SoR = {(a,x), (b,y), (b,z), (c,x)}

COMPOSITE RELATION FROM ARROW DIAGRAM:

Let A = {a,b,c},B = {1,2,3,4} and C = {x,y,z}. Define relation R from A to B and S from B to C by the following arrow diagram.





MATRIX REPRESENTATION OF COMPOSITE RELATION:

The matrix representation of the composite relation can be found using the Boolean product of the matrices for the relations. Thus if MR and MS are the matrices for relations R (from A to B) and S (from B to C), then

$$M_{SoR} = M_R \, O M_S$$

is the matrix for the composite relation SoR from A to C.

	BOOLEAN ADDITION		BOOLEAN MULTIPLICATION
a.	1 + 1 = 1	a.	1 . 1 = 1
b.	1 + 0 = 1	b.	1 . 0 = 0
c.	0 + 0 = 0	c.	0 . 0 = 0

EXERCISE:

Find the matrix representing the relations SoR and RoS where the matrices representing R and S are

$$M_R = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \qquad M_S = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 1 \end{bmatrix}$$

SOLUTION:

The matrix representation for SoR is

$$M_{SOR} = M_R O M_S = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

The matrix representation for RoS is

$$M_{ROS} = M_S O M_R = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 1 \end{bmatrix}$$

EXERCISE:

Let R and S be reflexive relations on a set A. Prove SoR is reflexive.